Cross-Border Spillovers of Bank Regulations: Evidence of a Trade Channel*

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Abstract

We study whether and how domestic banking regulations propagate internationally through trade linkages. We develop a stylized framework in which exporters respond to a credit-driven contraction in import demand through two margins: attenuating the shock by extending trade credit to constrained importers and reallocating exports across destinations. We test these mechanisms exploiting a large and unexpected increase in loan-loss provisioning requirements imposed on Spanish banks in 2012. Combining bank–firm credit registry data, confidential firm-level trade transactions, and global bilateral trade flows, we show that the regulation reduced credit supply to importers and led to a contraction in Spanish imports, generating spillovers to trading partners. Consistent with the model, exporters in financially developed economies—which are better positioned to extend trade credit—offset the shock, while exporters in less developed financial systems experienced larger declines, particularly when trade costs are high. Exporters of homogeneous products reallocate across markets, whereas differentiated products exhibit limited reallocation and larger export declines.

Keywords: Bank regulation; International trade; Cross-border spillovers; Trade credit; Financial development

JEL Codes: F14, F36, F42, G21, G28.

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1 Introduction

The Global Financial Crisis triggered a wave of regulatory reforms that significantly tightened banking regulation worldwide. While these reforms aim to enhance financial stability, they may also have generated unintended cross-border spillovers. Existing research has primarily examined *financial channels* of such spillovers, focusing on cross-border capital flows and the lending activities of global banks' foreign subsidiaries.[1] Much less attention has been paid to a distinct channel, one that operates even with limited capital market integration: The transmission of domestic financial shocks through an *international trade channel*.

This paper studies whether and how domestic banking regulations propagate internationally through trade linkages. We combine a stylized theoretical framework with micro-level credit and trade data to examine two adjustment margins: the use of trade credit to attenuate domestic financial shocks and the reallocation of trade flows across destinations. Exploiting a large and unexpected increase in loan-loss provisions imposed on Spanish banks in 2012, we show that tighter domestic regulation reduced import demand and generated significant cross-border spillovers, which are stronger when foreign exporters operate in financially underdeveloped economies, face higher trade costs, or produce differentiated goods that are difficult to redirect across markets.

A priori, it is unclear whether domestic banking regulations spill over through international trade. Domestic importers facing tighter credit supply may substitute toward alternative lending sources, such as trade finance provided by foreign exporters. If this substitution does not occur, less-affected domestic importers may capture the market share of more affected ones, leaving aggregate import demand largely unchanged. If, in turn, less-affected importers do not absorb the decline, foreign trade partners may still redirect exports toward alternative destinations.

To formalize these mechanisms, we develop a stylized model that focuses on foreign exporters' responses to a credit-driven contraction in domestic import demand. Exporters have two adjustment margins. First, they may extend trade credit to financially constrained importers, partially offsetting the domestic credit tightening. Second, they may reallocate output across destinations when demand from the shocked importer declines.

[1] For instance, Aiyar et al. (2014); Bahaj and Malherbe (2024); Buch and Goldberg (2017); Tripathy (2020).

In the model, importers rely on both domestic bank credit and foreign trade credit, with trade credit subject to partial recovery in the event of default: recovery rates decline with geographic distance and improve with prior relationships. The model delivers three key insights. First, trade finance usage declines with distance. Second, the import contraction driven by a domestic credit supply shock is most pronounced in distant markets, where substitution toward trade credit is limited by lower recovery rates, and in financially underdeveloped countries, where exporters' own financing capacity is limited. Third, prior relationships mitigate this contraction, with the attenuation effect strongest in markets where recovery frictions are most severe.

When the scope for trade-financing substitution is limited, foreign exporters can still adjust along a second margin by reallocating output across destinations. Exporters facing weaker demand from shocked importers can redirect goods to alternative markets, though this reallocation is subject to frictions such as repackaging costs, product adaptation, and regulatory barriers that make reallocation costly. These frictions imply that reallocation takes place only when the import contraction is large enough to justify the costs of redirecting exports, and when product characteristics facilitate such adjustment, e.g., homogeneous goods that can be reallocated with minimal adaptation.

To empirically assess these mechanisms, we exploit a large and unexpected increase in loan-loss provisioning requirements imposed on Spanish banks in 2012. The regulation targeted exposures to the construction and real estate sectors, non-tradable industries plausibly exogenous to global trade conditions, and generated substantial cross-sectional variation in banks' balance sheet exposure (Jiménez et al., 2017). Using bank–firm level data from the Spanish Central Credit Register, we show that more exposed banks reduced credit supply to importers after the reform. Exposed importers exhibit limited scope for substitution toward alternative domestic lenders or funding sources, and hence face a contraction of their overall financing.

We then combine the Spanish credit registry with transaction-level Balance of Payments data and global bilateral trade flows from BACI. Our analysis proceeds in three steps. First, we document that the regulation reduced firm-level imports among Spanish importers more exposed to affected banks. Specifically, using transaction-level data disaggregated by firm, product, country, and year,

and a difference-in-differences framework with rich fixed effects following Paravisini et al. (2014), we find that a one standard deviation increase in exposure reduces imports by 3.6 percent. Consistent with a domestic credit-supply mechanism, the effects are stronger for riskier importers, firms with weaker banking relationships, and those with limited access to credit lines, highlighting the role of bank ties in buffering adverse financial shocks (Bolton et al., 2016).[2]

Second, we turn to aggregate trade flows to assess whether firm-level contractions translate into cross-border spillovers. Using BACI bilateral trade data, we construct a measure of exposure to the policy at the product-origin level, capturing the extent to which Spanish importers of a given pair rely on funding from exposed banks. For example, if Spanish importers of textiles from India depend heavily on financing from highly exposed banks, then Indian textiles are highly exposed to the regulation. This level of aggregation is necessary to test for potential business stealing across Spanish importers, which could suppress any cross-border transmission (Matray et al. 2024; Beaumont et al. 2025).

Implementing a triple difference-in-differences strategy that compares exports to Spain from more- versus less-exposed product-origin pairs, relative to their exports to other destinations, we find little evidence of complete reallocation within Spain. A one standard deviation increase in exposure leads to a 2.1 percent decline in exports to Spain relative to other destinations of the same product and origin, with no evidence of pre-trends. These results remain robust when restricting the counterfactual destinations to comparable high-income countries or excluding GIIP exporters, addressing potential identification concerns related to the European sovereign debt crisis.[3] These findings indicate that the contraction in firm-level import demand translates into an aggregate decline in Spanish imports.

Third, we examine the two adjustment margins through which foreign exporters can cushion the shock: supply of additional trade financing and reallocation across destinations. In the model, exporters can mitigate a destination-specific credit-driven demand shock by extending trade credit to Spanish importers, but their ability to do so depends on their own financing capacity. We

[2]For completeness, we conduct a parallel analysis for domestic exporters. In contrast to importers, both exporter credit and firm-level export flows are only modestly affected by the regulation, with effects concentrated among smaller firms and negligible at the aggregate level.

[3]GIIP denotes Greece, Ireland, Italy, and Portugal.

therefore examine whether exporters operating in countries with more developed financial systems, proxied by private credit to GDP, are better positioned to absorb the shock.[4] Consistent with the model's attenuation mechanism, exporters in financially developed economies fully offset the credit-induced contraction, leaving their exports to Spain broadly unchanged. In contrast, exporters in less financially developed countries experience a significant 3.3 percent decline in exports to Spain.

Trade costs further shape this attenuation margin. Among exporters in less financially developed economies, where attenuation capacity is already constrained, the contraction is significantly stronger in distant and non-Spanish-speaking countries. A one standard deviation increase in exposure leads to a 3.6 percent decline in exports from distant countries and a 3.9 percent decline when distance and language frictions coincide. These aggregate estimates align closely with the model's prediction that distance-dependent trade-financing frictions amplify the cross-border transmission of domestic credit shocks.

We provide further evidence on this attenuation margin using importer-level micro-data. We estimate heterogeneous effects across Spanish importers with varying degrees of prior reliance on supplier credit. Firms with a higher pre-policy share of trade credit experience significantly smaller contractions in import flows, consistent with trade credit serving as an alternative funding source. Moreover, this attenuation effect is stronger when imports originate from higher trade friction origins, i.e., geographically distant partners or those without a common language or free trade agreement. This is consistent with the model's prediction that prior trade relationships are most valuable precisely when recovery constraints are more severe.

We next turn to the second adjustment margin emphasized in our framework: reallocation across destinations. Even if exporters cannot fully attenuate the contraction in Spanish demand through trade credit, they may mitigate its impact by redirecting sales toward alternative markets. The extent to which total exports decline therefore depends not only on exposure to the Spanish shock, but also on exporters' baseline dependence on Spain as a destination. To isolate the policy effect, we compare product-origin pairs with similar pre-shock shares of exports to Spain, ensuring that differences in exposure are not mechanically driven by initial trade dependence. On average,

[4] See, for example, Manova (2013) and Xu (2022) on the role of financial development in shaping trade outcomes.

we find limited evidence of systematic reallocation. Moreover, among product-origin pairs for which Spain represents a major export market, a one standard deviation increase in exposure leads to a significant 2.8 percent decline in total exports. This contraction is driven entirely by lower exports to Spain, with exports to other destinations remaining broadly unchanged, indicating limited scope for cross-market adjustment.

We then examine whether product characteristics shape the ability to reallocate. In our model, destination-specific demand conditions or regulatory requirements imply higher adjustment costs when redirecting differentiated goods than homogeneous ones. To capture this, we construct a measure of product heterogeneity based on cross-country price dispersion at the HS 2-digit level, classifying products in the top three quartiles as heterogeneous. For these products, total exports decline by 9 percent, entirely driven by a 16 percent contraction in exports to Spain. In contrast, homogeneous products exhibit no decline in total exports, as the contraction in exports to Spain is fully offset by increased sales elsewhere. These patterns are consistent with the reallocation margin in the model: when product-specific adjustment frictions are high, exporters face limited scope to redirect output across destinations, amplifying the cross-border transmission of the domestic demand shock.

Finally, because Spanish banks operate abroad and finance firms engaged in international trade, one concern is that our results may be confounded by their international lending activities. For instance, if Spanish banks are disproportionately active in financing firms that export to Spain, the estimated effects could reflect cross-border bank transmission rather than a trade-demand channel. To address this concern, we use BankFocus data to identify countries without a presence of Spanish banks and re-estimate our baseline specification restricting the sample accordingly. We find no meaningful deviation from our baseline results. Thus, our documented cross-border effects operate primarily through trade linkages and exporter-side financial conditions, rather than through the internal capital markets of Spanish global banks.

Overall, our findings show that tightening domestic financial regulation, even when targeted at local vulnerabilities, can generate significant cross-border real effects by contracting import demand. The magnitude of these spillovers depends on two distinct adjustment frictions. First, attenuation

frictions arise when exporters lack the financial capacity to extend trade credit, particularly in less financially developed economies and in environments with high trade-finance frictions. Second, reallocation frictions stem from product differentiation and limit exporters' ability to redirect sales of heterogeneous goods toward alternative destinations. These results highlight international trade as a central channel of regulatory spillovers and underscore the importance of financial development and cross-border coordination in prudential regulation to mitigate unintended disruptions to global trade.

Contribution to the literature. This paper contributes to four strands of literature. First, we extend the literature on the cross-border transmission of financial shocks. Traditional channels emphasize the role of global banks' internal capital markets in propagating funding shocks (Bruno and Shin, 2014; Hale et al., 2019; Bolton and Oehmke, 2018), bank regulations (Bahaj and Malherbe, 2024; Aiyar et al., 2014; Buch and Goldberg, 2017; Baskaya et al., 2017; Tripathy, 2020), and monetary policy (Cetorelli and Goldberg, 2012; Buch et al., 2019; Morais et al., 2019; Brauning and Ivashina, 2020). We identify a distinct trade-demand channel through which domestic banking regulation propagates internationally by contracting import demand. Unlike financial channels, this channel operates independently of domestic banks' foreign presence and depends on trade-linkage characteristics such as financial development in exporting countries and trade costs.

Second, we contribute to the literature on production networks and international input–output linkages (Carvalho et al., 2020; Costello, 2020; Alfaro et al., 2021; Huremovic et al., 2025; Burstein et al., 2008; Richmond, 2019; Di Giovanni and Hale, 2022; Brauning and Sheremirov, 2023; Auer et al., 2019; Boehm et al., 2019; Adelino et al., 2022; Ersahin et al., 2024; Romero, 2025). We show that international buyer–supplier relationships transmit domestic regulatory shocks abroad. The intensity of this transmission depends on exporters' ability to attenuate the shock through trade credit and on their capacity to reallocate across markets. In this sense, cross-border spillovers are shaped not only by financial linkages, but also by the structure of trade relationships.

Third, we contribute to the literature on export dynamics and cross-market interdependencies (Xu, 2022; Almunia et al., 2021; Jiao et al., 2024; Ding, 2024; Almeida et al., 2025). We show that when a financial shock reduces import demand in one market, exporters respond along two mar-

gins: attenuation through trade credit and reallocation toward alternative destinations. The scope for reallocation depends on product characteristics: exporters of homogeneous goods can redirect sales more easily, whereas differentiated goods exhibit more limited adjustment. Our analysis is particularly related to Xu (2022). We contribute by providing a mechanism that explains the heterogeneous cross-border effects of domestic financial shocks across markets, based on trade credit and product reallocation frictions. Moreover, we show that attenuation and reallocation margins operate in a modern regulatory context. Our findings also complement recent work documenting trade reallocation during periods of heightened uncertainty, such as trade wars (Alfaro and Chor, 2023; Benguria and Saffie, 2024; Gopinath et al., 2025), by highlighting the role of financial frictions in shaping cross-market export adjustment.

Fourth, we contribute to the extensive literature on trade finance (Amiti and Weinstein, 2011; Manova, 2013; Del Prete and Federico, 2014; Paravisini et al., 2014; Demir et al., 2017; Niepmann and Schmidt-Eisenlohr, 2017; Ahn and Sarmiento, 2019; Xu, 2022; Borchert et al., 2023; Moreira and Monteiro, 2023; Dogan and Hjortso, 2024; Matray et al., 2024; Beaumont and Lenoir, 2025). We advance this literature in two directions. First, we document a novel international spillover effect: tighter domestic credit conditions reduce foreign exporters' sales, particularly when exporters operate in financially underdeveloped economies and face higher trade costs—which limit their ability to extend trade credit—or produce goods that are difficult to reallocate. Second, we study both domestic imports and exports, showing that the cross-border effects of domestic credit shocks primarily operate through import demand rather than export supply.

While existing models show how financial frictions shape the organization of trade and the choice of financing contracts (e.g., Schmidt-Eisenlohr 2013; Antràs and Foley 2015; Niepmann and Schmidt-Eisenlohr 2017), we focus on the aggregate implications of trade credit relationships once these trading links are established. In our framework, exporters can attenuate foreign demand contractions by extending trade credit to their buyers, but the effectiveness of this margin depends on enforcement frictions that increase with distance and decline with repeated relationships. This setting delivers predictions on both the attenuation of export shocks and the reallocation of exports across markets.

The rest of the paper is organized as follows: Section 2 develops the theoretical framework. Section 3 provides institutional details of the policy intervention. Section 4 describes the data. Section 5 presents our main results and Section 6 discusses the mechanisms driving the cross-border spillovers. Section 7 concludes.

2 Conceptual Framework

We develop a stylized model to study how a domestic credit supply shock transmits across borders through international trade. The model illustrates two key margins of adjustment: foreign exporters can attenuate the contraction in import demand by extending trade credit, and they can reallocate output across destinations when attenuation is insufficient.

2.1 The Role of Trade Credit

Importers. There is a representative domestic importer for each home-country trade partner i. Importers purchase goods from abroad, M_i, to operate the production technology $y_i = z_i \ln M_i$, where z_i is the importer's productivity. A fraction τ_i of imports must be financed. Importers have two complementary sources of funding: bank credit L_i provided by the domestic banking sector and trade credit F_i provided by the foreign exporter.

The timing is as follows. In period $t = 0$, importers choose import value M_i and the financing structure L_i and F_i, taking interest rate schedules as given and without knowing their future productivity z_i. Instead, they know the distribution from which productivity will be drawn. In $t = 1$, productivity is realized and importers decide whether to default on trade credit or not. If they default, they are protected by limited liability and therefore obtain a normalized payoff of zero; if they repay, they receive $z_i \ln M_i - r_b L_i - r_x F_i$, where r_b and r_x denote the interest rates on bank credit and trade credit, respectively.[5]

At $t = 0$, taking interest rate schedules r_b and r_x as given, importer i maximizes expected

[5] We abstract from default on bank debt for simplicity. Instead, we assume that domestic banks fully recover defaulting loans. Additional results (not reported) show that allowing for selective default on both types of credit does not affect our conclusions.

profits:

$$\max_{\{M_i, L_i, F_i\}} \mathbb{E}\left[\max\left\{z_i \ln M_i - r_b L_i - r_x F_i,\ 0\right\}\right] \tag{1}$$

subject to the financing constraint

$$L_i + F_i = \tau_i M_i.$$

In $t = 1$, after observing z_i, importers decide whether to default on trade credit or not. Ex ante, the default probability on trade credit is given by:

$$\chi_i = \Pr\left(z_i \leq \frac{r_b L_i + r_x F_i}{\ln M_i}\right) \tag{2}$$

Domestic Banks. Each importer obtains credit from one bank who sets prices competitively. The bank supplies credit L to importers and faces no default risk. Banks fund loans with deposits D at a convex cost such that the competitive interest rate is:

$$r_b = r_{b,0} + \lambda_b L \tag{3}$$

where $r_{b,0}$ is the deposit rate and $\lambda_b > 0$ captures the increasing marginal cost of managing deposits.

Trade Credit. Foreign exporters provide trade credit and are exposed to default risk. In the event of default, which occurs with probability χ, the importer recovers a fraction α of the loan. Trade credit providers have a constant marginal cost $r_{x,0}$, which reflects financial development at exporters' country of origin. We further assume they do not compete with domestic banks; instead, they observe the value of imports and the amount already financed by domestic banks, and provide trade credit at the competitive interest rate:

$$r_x = r_{x,0} + \frac{\chi(r_{x,0} - \alpha)}{1 - \chi} \tag{4}$$

Solving the system of equations (2) and (4) yields r_x and χ as functions of the endogenous

choices L and F, and parameters $(r_{b,0}, \lambda_b, r_{x,0}, \alpha)$.

Functional Forms. We parameterize two key frictions. First, the fraction of imports requiring external financing increases with distance, capturing longer shipment times and greater working capital needs (Schmidt-Eisenlohr 2013; Xu 2022):

$$\tau_c = \beta_0 + \beta_1 \ln d_c. \tag{5}$$

Second, the recovery rate on trade credit declines with distance. To capture that recovery declines slowly at short distances but more rapidly once distance becomes large, we use a power function:

$$\alpha_c(\ell) = \overline{\alpha} - \delta(\ell) \times d_c^{\gamma}, \tag{6}$$

where $\overline{\alpha}$ is the maximum recovery rate, $\gamma > 1$ governs the curvature, and $\ell \in \{0, 1\}$ indicates whether a prior relationship exists.[6] We assume $\delta(1) < \delta(0)$, so relationships slow the rate at which recovery deteriorates with distance. This specification captures the idea that enforcement and monitoring costs rise non-linearly with distance (Schmidt-Eisenlohr 2013; Antràs and Foley 2015; de Nicola et al. 2026) and building relationships attenuates this friction (Petersen and Rajan 1994).

Comparative Static Analysis. We analyze the effect of a domestic bank credit supply shock by comparing two equilibria with different levels of $r_{b,0}$, the baseline cost of lending faced by domestic banks.

Figure 1 plots the equilibrium share of imports financed through trade credit against distance from the home market. The solid blue line represents importers with established supplier relationships prior to the shock; the dashed red line represents those without. While in both cases firms have access to trade credit, the expected recovery rate in default is lower in the absence of a relationship, which reduces equilibrium trade credit usage. Moreover, without a relationship, trade credit usage becomes more sensitive to distance, reflecting that recovery rates decline more steeply.

[6] The presence of a relationship is taken as given for the comparative static analysis.

Figure 2 illustrates how importers adjust their financing structure and total imports in response to a domestic credit supply shock. Panel (a) shows the change in trade credit usage, measured in percentage points, against distance. Firms rebalance toward trade credit, but the magnitude of this adjustment declines with distance, and more sharply so in the absence of a prior relationship. A relationship is not a prerequisite for accessing trade credit; it instead mitigates frictions that affect recovery rates (Petersen and Rajan 1994). Without it, distance sensitivity is amplified, leaving importers in more distant markets less able to substitute away from domestic credit when local conditions tighten.

Panel (b) reports the corresponding decline in total imports by distance. Consistent with the limited scope for financing rebalancing in distant markets, the contraction in imports is more pronounced as distance increases, particularly when no prior relationship exists.

The limited substitution as distance increases stems from financial frictions in recovery rates, which generate steeper marginal costs of switching to trade credit in more distant markets—especially without prior relationships. To illustrate this mechanism, Figure 3 presents a partial equilibrium analysis of an increase in $r_{b,0}$, holding total imports M fixed. Panel (a) considers a close market, plotting the marginal cost of domestic credit before and after the shock. The marginal cost of trade credit is relatively flat, reflecting a high recovery rate, and varies little when there is no prior lending relationship. As a result, following the domestic shock, importers can easily substitute toward trade credit, and the equilibrium marginal cost rises modestly.

Panel (b) considers a distant market, revealing a key difference: the marginal cost of trade credit now depends heavily on whether a prior relationship exists. In its absence, the marginal cost curve is substantially steeper. Consequently, importers shift slightly toward trade credit and face a sharp increase in the equilibrium marginal cost.

Our model yields several testable implications summarized in the following propositions.

Proposition 1. *The share of imports financed through foreign trade credit declines with distance between trading partners. For any given distance, importers with prior relationships rely more on trade credit than those without.*

Proof. See Appendix A.

Proposition 2. *The decline in imports induced by an increase in the baseline domestic lending rate $r_{b,0}$ is attenuated when recovery rates on trade credit are higher:*

$$\frac{\partial^2 \ln M}{\partial \alpha \, \partial r_{b,0}} > 0.$$

Proof. See Appendix A.

Higher recovery rates reduce the marginal cost of trade credit and therefore facilitate substitution away from domestic bank credit when local lending conditions tighten. Because recovery rates decline with distance, the scope for substituting toward trade credit becomes more limited in distant markets, leading to larger import contractions following a domestic credit shock. Prior relationships mitigate these enforcement frictions by improving recovery rates. Since the recovery advantage associated with relationships increases with distance, the attenuation effect of relationships is amplified in more distant markets.

2.2 The Role of Export Reallocation

We now examine how foreign exporters' ability to reallocate exports shapes the cross-border transmission of domestic bank regulations, taking the import decline as given and abstracting from the trade credit decision. We use a two-period model to evaluate this reallocation margin.

Exporters. Consider a foreign exporter serving two countries, $i = 1, 2$. In each country, the exporter faces linear demand $p_i = a_i - b_i q_i$, where q_i is quantity sold and p_i the local price. The timing is as follows. In period $t = 1$, exporters choose quantities for each destination, observing the demand functions. To deliver q_i units to country i, the exporter must produce $(1 + \tau)q_i$ units, where τ is an iceberg trade cost. We assume linear production costs such that the total cost of delivering q_i units to country i is $(1 + \tau)q_i$. In period $t = 2$, goods are shipped and revenues is collected. Thus, in period $t = 1$, exporters solve:

$$\max_{q_1, q_2} \sum_{i=1}^{2} \left[(a_i - b_i q_i) q_i - (1 + \tau) q_i \right]. \tag{7}$$

The optimal choice is given by:

$$q_i^* = \frac{a_i - (1+\tau)}{2b_i}, \qquad i = 1, 2 \tag{8}$$

which represent the planned quantities q_1^* and q_2^* the exporter intends to sell in each country. To achieve these sales, the exporter produces in $t = 1$ $(1+\tau)q_i^*$ physical units for country i.

Import demand shock. Between periods, an unexpected negative demand shock hits country 1, reducing a_1 to $a_1' < a_1$. Let $\Delta a = a_1 - a_1' > 0$ denote the size of the shock. The exporter has already produced $(1+\tau)q_i^*$ physical units for each country and must now decide how to allocate them. It can redirect Δ physical units originally destined for country 1 to country 2, but doing so entails two costs. First, a fixed cost $f > 0$ must be paid to undertake reallocation (e.g., administrative expenses, new shipping arrangements). Second, a fraction $\gamma \in (0, 1)$ of the redirected units is lost during the process, capturing frictions such as repackaging, relabeling, or logistical re-routing required to make goods suitable for the alternative destination.

After redirection, the physical units available for shipment to country 1 are $(1+\tau)q_1^* - \Delta$, and for country 2 they are $(1+\tau)q_2^* + (1-\gamma)\Delta$. When these units are shipped, the iceberg trade cost applies: only a fraction $1/(1+\tau)$ of shipped units actually arrive at their destination. Defining $\delta = \Delta/(1+\tau)$ as the reduction in sales to country 1, the final quantities sold become $q_1 = q_1^* - \delta$ and $q_2 = q_2^* + (1-\gamma)\delta$.

No reallocation. If the exporter does not reallocate, it can still choose how many units to sell in country 1, up to the available stock q_1^*. Let q_1 be the quantity sold; the remainder $q_1^* - q_1$ is destroyed to support a higher price. Revenue from country 1 is $R_1(q_1) = (a_1' - b_1 q_1)q_1$, which is concave and maximized at $a_1'/(2b_1)$. Because sales cannot exceed the available stock, the optimal quantity is $q_1^{\text{opt}} = \min\left(q_1^*, \frac{a_1'}{2b_1}\right)$. Then, revenue in country 1 is as follows:

$$R_1^{\text{no}}(\Delta a) = \begin{cases} (a_1' - b_1 q_1^*)q_1^*, & \Delta a \leq 1+\tau, \\ \dfrac{(a_1')^2}{4b_1}, & \Delta a > 1+\tau \end{cases}$$

Country 2 is unaffected, so its revenue remains $R_2^* = (a_2 - b_2 q_2^*)q_2^*$. Total profit without reallocation is:

$$\Pi_{\text{no}}(\Delta a) = R_1^{\text{no}}(\Delta a) + R_2^* \tag{9}$$

Reallocation decision. If the exporter decides to reallocate, it pays the fixed cost f and chooses $\delta \in [0, q_1^*]$ to maximize revenue after redirection:

$$\max_{\delta \in [0, q_1^*]} R(\delta) - f, \tag{10}$$

where

$$R(\delta) = \big[a_1' - b_1(q_1^* - \delta)\big](q_1^* - \delta) + \big[a_2 - b_2(q_2^* + (1-\gamma)\delta)\big]\big(q_2^* + (1-\gamma)\delta\big).$$

The first-order condition for an interior optimum yields the optimal reallocation quantity:

$$\delta^* = \frac{\Delta a - \gamma(1+\tau)}{2\big[b_1 + b_2(1-\gamma)^2\big]}, \tag{11}$$

If $\Delta a < \gamma(1+\tau)$, the numerator is negative and the optimum is $\delta = 0$, i.e., reallocation is not beneficial. If $\delta^* > q_1^*$, exiting country 1 is optimal. Note that the fixed cost f does not affect the choice of δ^* conditional on reallocating.

Proposition 3. *Following a negative demand shock in one destination, exporters find it optimal to reallocate a positive quantity toward alternative markets only when the shock size exceeds $\gamma(1+\tau)$. The magnitude of reallocation increases with the shock size and, for a wide range of parameters, decreases with the friction γ.*

Proof. See Appendix A.

Optimal profit with reallocation is therefore $\Pi_{\text{re}}(\Delta a) = R^*(\Delta a) - f$, where $R^*(\Delta a)$ denotes the maximized revenue. When deciding whether to reallocate, the exporter weighs this against the profit from no reallocation, $\Pi_{\text{no}}(\Delta a)$. Defining the gain from reallocation as $G(\Delta a) = \Pi_{\text{re}}(\Delta a) - \Pi_{\text{no}}(\Delta a)$, we have the cutoff rule characterized in Proposition 4.

Proposition 4. *There exists a unique threshold $\Delta a^* > \gamma(1+\tau)$ such that $G(\Delta a^*) = 0$, with reallocation occurring iff $\Delta a > \Delta a^*$. This threshold increases with the fixed cost f.*

Proof. See Appendix A.

This framework yields two main implications. First, the fixed cost f creates a threshold effect: reallocation occurs only when the shock size exceeds a critical value, which increases with f. Second, the redirection friction γ dampens reallocation: products facing higher redirection costs exhibit smaller δ^* and consequently larger overall export declines following a negative demand shock in country 1.

This reallocation mechanism complements the trade financing channel developed earlier. Trade credit allows foreign exporters to attenuate the shock by extending financing to importers. However, when collateral frictions limit exporters' ability to provide trade credit, the resulting import decline may be large enough to trigger reallocation toward unaffected markets.

3 Empirical Setting

Guided by the theoretical framework developed in Section 2, we now bring the model's predictions to the data. The framework highlights two margins through which a domestic credit supply shock can propagate internationally: an attenuation margin, whereby foreign exporters cushion the contraction in import demand by extending trade credit, and a reallocation margin, whereby exporters redirect sales toward alternative destinations.

Empirically identifying these mechanisms requires isolating variation in domestic credit supply that is orthogonal to foreign demand or supply shocks. We exploit the 2012 Spanish provisioning reform as an exogenous shock to bank lending conditions. Because the reform targeted pre-existing exposures to construction and real estate, non-tradable sectors plausibly unrelated to global trade

dynamics, it generates cross-sectional variation in banks' balance sheet exposure that is independent of contemporaneous foreign developments.

3.1 The 2012 Spanish Provisioning Reform

In early 2012, the Spanish government imposed a one-time, two-phase increase in provisioning requirements for banks' exposures to the construction and real estate sectors, based on outstanding loans as of late 2011. The policy aimed to reduce uncertainty around asset valuations still distressed after the 2008 Global Financial Crisis.

In the first phase (February), banks were required to provision 7% for all performing loans in the targeted sectors. In the second phase (May), provisioning was differentiated by loan type: an additional 45% for land, 7% for completed properties, and 22% for real estate projects in progress. Provisions for non-performing loans were also raised, and all requirements had to be met by year-end 2012.

This ad hoc policy represented a substantial shock: banks' provision buffers had already been depleted by the global crisis, leaving them unable to meet the new requirements without sharply increasing provisions by approximately $85 billion (8.5% of GDP), severely affecting bank profitability in 2012 (Jiménez et al., 2017).

Following Jiménez et al. (2017), we measure banks' exposure to the policy using the share of their corporate lending to construction and real estate firms before the reform. While their analysis documents a reduction in credit supply to corporate borrowers by highly exposed banks, we focus on the implications for non-financial firms engaged in international trade, and the resulting effects on trade flows and cross-border spillovers.

Among the 83 banking groups lending to exporters and importers, exposure to construction and real estate sectors varied widely. At the 25th, 50th, and 75th percentiles, banks allocated 26.5%, 34.9%, and 47.0% of corporate credit to these sectors, respectively (see Panel C of Table 1). Figure 4 illustrates the distribution of bank exposure.

4 Data

Our analysis relies on three primary datasets: (i) the Spanish Central Credit Register (CIR) from the Bank of Spain; (ii) Spain's Balance of Payments data, used to compile the Bank of Spain's official statistics; and (iii) international trade data from BACI (Gaulier and Zignago, 2010), which provides comprehensive bilateral trade flows at the country-product level.

4.1 Credit Register (CIR)

To examine the bank lending channel of the regulation, we use quarterly bank-firm credit data from the CIR, a comprehensive registry from the Bank of Spain that covers all loan commitments exceeding €6,000 granted to non-financial firms by any credit institution operating in Spain, including loan type, drawn and undrawn amounts, collateral status, maturity, currency, past-due status, lender and borrower identities, and firm-specific characteristics such as industry classification and geographic location. To enhance our analysis, we combine CIR data with supervisory bank balance sheets and Balance of Payments data. This allows us to capture bank characteristics alongside firms' import and export activities and to link the bank supplying the credit with its exposure to the policy, captured by its share of credit granted to the construction and real estate industries. Our dataset includes credit information for 33,361 non-financial firms engaged in international trade, provided by 83 banking institutions from 2011Q1 to 2013Q2, yielding a total of 1,288,007 observations.[7] Descriptive statistics for the variables used in this analysis are presented in Panel A of Table 1, while variable definitions can be found in Appendix C.

4.2 Balance of Payments Data

Our analysis of firms' international trade outcomes relies on confidential transaction-level import and export records. This administrative dataset is one of the sources used by the Bank of Spain to compile Spain's official Balance of Payments. It captures the majority of trade flows in and out of Spain and provides detailed information for each importer and exporter, including the product code

[7] We control for bank mergers by considering the industry structure as of the end of our period of analysis. That is, we assume that new consolidated banks operate as a single entity within our period of analysis.

(based on 2-digit Harmonized System (HS) codes), the trading partner country, and the transaction year.[8]

Our analysis covers 2009 to 2013, during which all transactions exceeding €50,000 were subject to mandatory reporting. Despite this threshold, the dataset captures 91.3% of total trade flows relative to official Customs statistics (Almunia et al., 2021; Prades and Villegas-Sánchez, 2026; Gutiérrez and Moral-Benito, 2024).

The dataset covers 558,540 import observations (23,029 firms) and 649,050 export observations (25,173 firms) across 99 products, 167 import origins, and 189 export destinations (Panel C of Table 1).

4.3 BACI Data

To assess international spillover effects, we use international trade data from BACI (Gaulier and Zignago, 2010), an annual panel of bilateral trade flows at the origin-destination-product level, which we merge with a measure of exposure to the policy at the product–country-of-origin level, constructed using Balance of Payments and CIR data.[9] To enrich our dataset, we incorporate information from Moody's Analytics BankFocus, which provides annual balance sheet data on over 38,000 banks worldwide, allowing us to assess the presence of Spanish bank subsidiaries in the banking systems of Spain's trade partners. Additionally, we use the CEPII Gravity database, which offers country-pair-specific characteristics that shape trade costs, such as geographical distance, trade agreements, and common language.

Our trade data include 3,141,210 trade flow observations between 190 countries (origin and destination) and 5,216 HS 6-digit products grouped in 96 HS 2-digit goods between 2009 and 2013 (see Panel D of Table 1).

[8] Balance of Payments transactional data are reported either by banks on behalf of their clients or directly by firms to the Bank of Spain if they hold their own accounts.

[9] For the years 2005–2013, Spanish import flows obtained from Balance of Payments data at the 2-digit product-country level represent approximately 90% of BACI data.

5 Domestic Credit Shock and Import Contraction

In this section we present our first set of results. We begin by documenting the domestic effects of the reform on bank lending and firm-level imports. We then turn to aggregate trade data to assess whether these firm-level effects translated into cross-border spillovers, testing for business stealing across differently treated Spanish importers.

5.1 Domestic Credit Contraction

We begin by verifying that the 2012 provisioning reform generated a contraction in domestic credit supply to domestic importers. Using quarterly bank–firm data from the CIR, we estimate whether banks more exposed to the regulation reduced lending differentially after its implementation.[10]

Specifically, over the period 2011Q1–2013Q2, we estimate:

$$\log y_{f,b,t} = \theta \times Exposure_b \times Policy_t + controls_{b,f,t} + \alpha_{f,t} + \alpha_{f,b} + \varepsilon_{f,b,t}, \tag{12}$$

where $y_{f,b,t}$ denotes committed credit from bank b to firm f in quarter t. $Exposure_b$ measures bank b's pre-reform share of lending to construction and real estate (as of the end of 2011), and $Policy_t$ equals one from 2012Q1 onward. $Controls_{b,f,t}$ account for various bank and bank-firm characteristics measured as of the end of 2011 interacted with time fixed effects. Specifically, bank–firm controls comprise the collateralization rate, the share of long-term lending, the fraction of firm f's non-performing loans at bank b, and bank b's share in the firm's total borrowing. Bank-level controls consist of indicator variables for quartiles of bank size (by total assets) and quartiles of trade-sector exposure, measured as the share of lending to exporters and importers in total corporate lending. Importantly, our balance-of-covariates analysis indicates that, after controlling for these bank bins, the bank exposure measure is statistically orthogonal to other key bank characteristics, such as ROA, NPLs, liquidity, capital ratios, and credit to the public sector divided by total assets (see Figure 5).[11]

[10] Our analysis excludes firms operating in the construction and real estate sectors.

[11] This last check addresses the concern that our exposure measure might proxy for banks' sovereign debt holdings, and thus reflect differential exposure to the sovereign debt crisis rather than the provisioning shock. Public informa-

Firm–time fixed effects absorb credit demand shocks, so identification stems from within-firm variation across banks with differential exposure (Khwaja and Mian, 2008).[12] Exposure is standardized, so θ captures the effect of a one-standard-deviation increase.

Table 2 presents the bank–firm level estimates for domestic importers in columns (1)–(2). More exposed banks significantly reduced credit supply following the reform. A one standard deviation increase in bank exposure is associated with a 3.1 percent decline in committed credit relative to other lenders to the same firm. The contraction operates along both the intensive and extensive margins: exposed banks reduced credit volumes and were more likely to terminate lending relationships. Event-study estimates in Figure 6 (Panel A) show no evidence of differential pre-trends and a sharp contraction beginning in 2012, consistent with the timing of the regulatory shock.

For completeness, columns (3)–(4) report the analogous estimates for domestic exporters, and Panel B presents the corresponding event-study, which displays estimates of similar magnitude to those of importers.

We next assess whether importers were able to offset this shock by reallocating borrowing across banks. We estimate:

$$\log y_{f,t} = \theta \times Exposure_f \times Policy_t + controls_{f,t} + \hat{\alpha}_{f,t} + \alpha_f + \varepsilon_{f,t}, \tag{13}$$

where $y_{f,t}$ denotes total bank credit to firm f, and $Exposure_f$ is the credit-weighted average exposure of its lenders prior to the reform. As controls, we interact quarter dummies with firm-level characteristics constructed as credit-weighted averages of the bank–firm variables used in specification (12), weighted by each bank's outstanding credit to the firm at end-2011. We also include province and industry indicators. To account for firm-level demand shocks, we incorporate the estimated firm–quarter fixed effects from equation (12), $\hat{\alpha}_{f,t}$, following Abowd et al. (1999) and Bonaccorsi di Patti and Sette (2016).

tion released as part of the European Banking Authority's 2011 stress test indicates that Spanish banks' sovereign exposures were heavily home-biased and held predominantly in the banking book. A proxy to the banking book, given by the share of public sector credit to total assets, computed using CIRBE, is shown to be uncorrelated with our measure of bank regulatory exposure.

[12] Firms engaged in international trade in our sample typically maintain multiple banking relationships. In 2011, only about 30% of trading firms borrowing from banks relied on a single lender.

Table 3 shows that importers more exposed to affected banks experienced a 2.8 percent decline in total bank debt, indicating limited substitution toward less exposed lenders. Again, we find no evidence of differential pre-trends (Figure 7, Panel A).

Moreover, Table 4 shows no evidence of substitution toward non-bank financing (columns (2)-(3)): importers more exposed to affected banks experience declines not only in total bank debt but also in total debt and total debt plus equity, indicating limited access to alternative funding sources. The contraction in credit is also accompanied by negative real effects—reductions in employment and tangible assets—reinforcing the economic significance of the shock (columns (4)-(5)).

For completeness, the corresponding estimates for exporters indicate substantially smaller effects, suggesting that exporters were better able to rely on multiple bank and non-bank relationships to partially cushion the shock.[13]

Appendix B presents additional robustness tests showing that our results are not driven by bank specialization in foreign markets (Table B.2) or foreign demand shocks (Table B.3).

Having established that the reform tightened credit conditions for importers, we next examine whether this contraction translated into reduced import demand and how trade credit relationships shaped it.

5.2 Effect on Firm-Level Trade Flows

Using confidential Balance of Payments data, we observe firm-level trade flows by partner country, 2-digit product, and year. This granularity allows us to estimate the impact of the policy while flexibly controlling for foreign supply shocks and local demand conditions.

We estimate the following specification:

$$y_{f,c,g,t} = \alpha_{f,c,g} + \alpha_{c,g,t} + \alpha_{p(f),i(f),t} + \theta \times Exposure_f \times Policy_t + \varepsilon_{f,c,g,t}, \tag{14}$$

[13] Our baseline importer–exporter classification allows for overlap. Table B.1 in the Appendix separates firms that are only importers, only exporters, or both, and confirms that exclusively importing firms experience the largest credit declines, whereas exclusively exporting firms better cushion the shock through multiple banking relationships. Moreover, Table B.4 and Table B.5 show that exporters are on average larger and financially stronger than importers, with greater debt capacity and more stable funding structures.

where $y_{f,c,g,t}$ denotes the mid-point growth rate of firm f's imports of product g from country c in year t, relative to its pre-policy average.[14] This specification captures both intensive and extensive margins, as zero trade flows reflect market entry and exit (Fonseca and Matray, 2024; Matray et al., 2024; Beaumont et al., 2025). $Exposure_f$ is defined as in equation (13), and $Policy_t$ equals one in 2012–2013. Firm–country–product fixed effects absorb time-invariant relationship characteristics, while country–product–year fixed effects control for foreign supply shocks. Province–industry–year fixed effects account for local economic conditions.[15]

Table 5 reports the results. A one standard deviation increase in exposure to affected banks is associated with a 3.6 percent decline in imports following the reform (column 1), roughly 14 percent of the year-on-year import growth rate in the year prior to the policy. The contraction operates along both margins: more exposed firms are significantly more likely to exit import markets and less likely to enter new ones. Event-study estimates confirm no pre-trends and a sharp post-reform decline in import growth (see Figure 8).

We further examine whether the import contraction varies with firms' financial conditions, as implied by a credit-driven mechanism. Consistent with this interpretation, Table 6 shows that importers more reliant on a single relationship bank experience smaller trade contractions, as relationship lenders tend to shield clients during downturns (Bolton et al., 2016). Similarly, firms with greater pre-existing credit line availability exhibit weaker import declines, while those with higher non-performing loan ratios experience sharper contractions. These patterns confirm that liquidity buffers and strong banking relationships mitigate the shock, while weaker balance sheets amplify it.

Appendix B reports additional robustness tests confirming that the results are not driven by specific trading partners or product categories (Figure B.1, Figure B.2), firms' exposure to countries affected by the European sovereign debt crisis (Table B.6), or sectors indirectly linked to

[14] Specifically,

$$y_{f,c,g,t} = \frac{Z_{f,c,g,t} - \overline{Z}_{f,c,g}}{1/2(Z_{f,c,g,t} + \overline{Z}_{f,c,g})},$$

where $\overline{Z}_{f,c,g} = 1/5 \sum_{t=2009}^{2013} Z_{f,c,g,t}$ for $Z =$ Imports, Exports.

[15] Given the granularity of our data, we exclude transactions representing less than 1% of total import (35,000 euros) or export value (29,000 euros) over 2009–2013, to avoid results driven by very small or highly volatile flows.

construction and real estate through input-output linkages (Table B.7).

For completeness, we report the corresponding trade flow estimates for domestic exporters, which are consistently weaker and less robust than those for importers, in line with our credit results. In Appendix B, we further estimate trade-weighted regressions (Table B.8), which render the export effects statistically insignificant, while the impact on imports remains robust. In addition, Table B.9 shows that the export response is concentrated among small exporters (bottom 25% by export value), with no significant effect for larger firms that account for the majority of aggregate exports.[16] By contrast, the contraction in imports is broad-based across firm size groups.

5.3 Aggregate Cross-Border Effects

We estimate the extent to which the increase in Spanish banks' loan loss provisions affected Spain's trade partners through international trade linkages. As previously noted, the decline in imports among Spanish firms highly exposed to the policy—through their relationships with the most affected banks—does not necessarily imply a contraction in Spain's aggregate imports. For example, reduced demand for machinery from highly exposed firms in the wine industry could, in equilibrium, be offset by increased demand from less exposed firms whose banks were less affected by the policy.[17] Thus, a small or even null response in total Spanish imports would still be consistent with our firm-level findings.

To test the role of these general equilibrium forces, we use annual BACI data covering 2009–2013 and construct a measure of exposure to the policy at the product–country-of-origin level:

$$\text{Exposure}_{go} = \sum_b w_b^{go} \times \text{Exposure}_b \,, \quad \text{where} \quad w_b^{go} = \frac{\sum_f L_{fb} \times M_f^{go}}{\sum_b \sum_f L_{fb} \times M_f^{go}} \tag{15}$$

[16] A potential concern is that these findings are confounded by the export boom documented by Almunia et al. (2021), who show that post-crisis competitiveness gains led to a 31% increase in Spanish exports between 2009 and 2013. We view this as unlikely for two reasons. First, our specification includes province- and industry-year fixed effects that absorb the local shocks highlighted in their analysis. Second, we find no evidence of geographic sorting in our bank exposure measure: provinces experiencing larger domestic sales declines were not disproportionately served by banks more exposed to the regulation (Figure B.3).

[17] Business stealing is particularly relevant for firms engaged in international trade. Since they are typically large, they can capture the market share from competitors facing negative shocks (Matray et al. 2024; Beaumont et al. 2025).

and L_{fb} represents the outstanding debt that firm f holds with bank b, and M_f^{go} denotes firm f's imports of product g from country o. This measure represents the weighted average exposure of banks, where the weights w_b^{go} capture each bank's relative importance in financing imports of good g from country o. To construct this measure, we use information from the end of 2006–three years prior to the start of our sample period.[18] A product–country-of-origin pair is considered highly exposed to the policy if exports from that pair to Spain are primarily purchased by importers relying more heavily on banks most exposed to the regulation. For example, if Spanish importers of textiles from India depend heavily on financing from highly exposed banks, then Indian textiles are highly exposed to the regulation. The distribution of this exposure measure, shown in Figure 9, reveals substantial variation across pairs.

We begin by estimating the regulation's impact on Spain's total imports (equivalently, other countries' exports to Spain). Specifically, we compare how exports to Spain evolved across product-origin pairs with different exposure levels, relative to the exports from the same pairs to other destinations, before and after the policy. Formally, we implement a triple difference-in-differences design:

$$\frac{X_{godt} - \overline{X}_{god}}{(X_{godt} + \overline{X}_{god})/2} = \gamma \times \text{Exposure}_{go} \times \mathbb{1}[d = \text{Spain}] \times Post_t + \delta_{god} + \delta_{gt} + \delta_{ot} + \delta_{dt} + u_{godt}, \quad (16)$$

where X_{godt} denotes exports of product g from country-of-origin o to country-of-destination d in year t. Thus, our dependent variable is a mid-point growth rate relative to the corresponding average value in the sample period. $Exposure_{go}$ is defined as in equation (15) and is multiplied by an indicator equal to one when Spain is the importing country and by $Post_t$, which is a dummy that takes the value of one in the years after the policy implementation (2012 and 2013). We include time-invariant product×origin×destination fixed effects denoted by δ_{god}, and time-varying fixed effects at the product, origin, and destination level, denoted by δ_{gt}, δ_{ot}, and δ_{dt}, respectively. Given the granularity of our data, we trim extreme values of Spanish participation by excluding

[18] From 2008 onward, the representativeness of the Balance of Payments data by country and product declines due to a change in reporting procedures; see Prades and Villegas-Sánchez (2026) for details. We therefore use 2006 as the baseline year to compute foreign exporters' exposure and match it with BACI trade flows. For all other exposure measures–bank-level and firm-level–we employ end-of-year 2011 data.

product-country pairs where Spanish purchases represent less than 1.1 percent (the median). After this trimming, the sample covers 91 percent of Spain's total purchases.

Our parameter of interest, γ, captures the impact of Spanish loan loss provisions on exports to Spain relative to exports to other countries. If there is business stealing across Spanish firms, total exports to Spain will not differentially respond to our product–country-of-origin exposure after the policy, and γ will be zero. We identify γ using a triple-differences approach with time-varying product, origin, and destination fixed effects, which controls for product- and origin-specific trends coinciding with the increase in loan loss provisions as well as for demand shocks occurring simultaneously with the policy. Thus, our identifying assumption is that, absent the policy, the difference in exports to Spain from high- relative to low-exposed country-product pairs would have evolved similarly to the corresponding difference in other importing countries.

As a robustness, we re-estimate the triple-differences specification replacing destination×year fixed effects with a restricted set of importers comparable to Spain. Specifically, we consider EU members and other advanced economies as defined by Autor et al. (2013).[19] Additionally, we exclude Greece, Italy, Ireland, and Portugal (GIIP) from the set of exporters to address potential identification concerns related to the decline in their exports to Spain during the European sovereign debt crisis.

We report our results in Table 7. Column (1) shows that a one-standard-deviation increase in exposure is associated with a 3.2 percent decline in exports to Spain relative to exports to other countries in a naive specification with exporter-destination and destination-year fixed effects. Column (2) reports a similar effect in a more saturated specification that includes time-varying product fixed effects. Our benchmark specification in Column (3) adds time-varying origin fixed effects to control for supply shocks and indicates that a one standard deviation increase in product-origin exposure reduces exports to Spain by 2.1 percent after the policy, relative to the same product-origin pair's exports to other destinations in the same period. Column (4) shows similar results when restricting the sample of destination countries to EU founders and other advanced

[19] This is, high-income countries with comparable trade data: Australia, Belgium, Denmark, Finland, France, Germany, Greece, Ireland, Italy, Japan, Luxembourg, Netherlands, New Zealand, Portugal, Spain, Switzerland, and the United States.

economies that are likely to face demand patterns comparable to Spain. Finally, Column (5) finds consistent effects when excluding GIIP exporters from the sample.

Figure 10 presents the event-study corresponding to our benchmark specification. Prior to the policy, exposure has no significant effect on exports to Spain, consistent with our identifying assumption; after the policy, exports to Spain from highly exposed origin-product pairs decline persistently relative to other destinations. Overall, these results confirm that business stealing across Spanish importing firms was limited: the firm-level contraction documented in Section 5.2 translated into an aggregate import decline.

Despite the advantages of our triple-differences specification in controlling for multiple demand and supply shocks beyond the increase in Spanish loan loss provisions, the narrow level of disaggregation in our data raises the concern that results may be disproportionately driven by small destinations (i.e. partners with negligible trade volumes), which could follow different dynamics than larger ones. Furthermore, analyzing the attenuation and reallocation margins of adjustment in the following subsections requires aggregating the data to the exporter-product level.

Thus, we estimate the following standard difference-in-differences specification separately for exports to Spain and to the rest of the world, to validate its consistency with our triple-differences estimation:[20]

$$\frac{X_{got} - \overline{X}_{go}}{(X_{got} + \overline{X}_{go})/2} = \beta \times \text{Exposure}_{go} \times Post_t + \delta_{go} + \delta_{x(g)t} + \delta_{x(o)t} + \delta_{x(o,g)t} + u_{got}, \tag{17}$$

where X_{got} denotes exports of product g from country o either to Spain or the rest of the world in year t. In this aggregate analysis, we cannot saturate the model with high-dimensional fixed effects. Instead, we include time-varying fixed effects for different categories of exporters and products: $\delta_{x(o)t}$ captures exporter characteristics such as distance to Spain and official language, while $\delta_{x(g)t}$ accounts for product differentiation.[21] Finally, we include product–country-of-origin level time-varying fixed effects by deciles of Spanish participation, $\delta_{x(o,g)t}$ to control for Spain's

[20] This latter specification acts as a placebo test, since the credit shock originated in Spain and should not systematically affect exports to other destinations.

[21] As we explain below, we build quartiles of price dispersion within product categories to measure product differentiation.

relative importance in a given market.

Our results are consistent with the triple difference-in-differences estimates. The benchmark specification reported in column (3) of Table 8 indicates a 2.7 percent decline in exports to Spain. Column (7) serves as a placebo: the response of exports to other countries is insignificant, confirming the shock was specific to Spain. Results remain unchanged when excluding GIIP exporters (columns (4) and (8)). The event-study in Figure 11 shows a post-reform decline with no pre-trends. Overall, results are consistent with those in Table 7.

6 Mechanisms of Cross-Border Transmission

Having established that Spanish import demand declined as a result of the policy, we now explore how foreign exporters (Spain's trade partners) responded to this contraction. Guided by our stylized model, we examine two adjustment margins: attenuation via trade finance, and reallocation across destinations.

6.1 Attenuation via Trade Credit

Exporters can attenuate the credit-driven decline in Spanish demand by offering more favorable trade finance terms to Spanish importers. However, exporters' capability to offer trade credit relies on the degree of financial development at origin, as indicated by our model. We test the hypothesis that exporters with access to more developed financial sectors are more likely to offer favorable trade finance terms.[22]

Our results in Table 9, where we estimate equation (17) with exports to Spain as the dependent variable, show how financial development shapes the attenuation of the decline in Spanish demand. Column (1) shows that exporters in highly financially developed countries–defined as those in the top quartile of the private credit-to-GDP ratio distribution–fully offset the Spanish credit contraction, leaving exports to Spain unaffected. By contrast, column (2) indicates that a one standard deviation increase in product-origin exposure reduces exports to Spain by 3.3 percent

[22] It is well established that countries with more developed financial markets export more, particularly in industries that rely heavily on external financing (Beck 2002, 2003).

when exporters belong to countries with weaker financial development.[23] These results support our hypothesis that exporters in deeper financial systems are better able to absorb credit-driven demand shocks by extending more favorable trade finance conditions to importers. This evidence underscores financial development as a key factor in containing cross-border spillovers of bank regulatory shocks through international trade linkages.

Then, we explore how country-specific characteristics that are associated with trade-finance frictions interact with financial development. For instance, exporters located farther from Spain face higher monitoring and verification costs, in addition to higher variable trade costs as longer shipment times intensify liquidity needs. Likewise, exporters from non-Spanish-speaking countries may face higher trade-financing frictions, beyond higher fixed trade costs as exporting to Spain often requires investments to overcome language barriers.

Our results are reported in Table 9. The decline in exports from less financially developed countries is concentrated among those that are geographically distant from Spain and do not share a common language. Columns (3) and (4) show that a one standard deviation increase in exposure leads to a statistically insignificant 2.2 percent decline for exporters located close to Spain (in the bottom quartile of the distance distribution), but a significant 3.6 percent decline for those farther away. Similarly, columns (5) and (6) indicate no effect for Spanish-speaking countries, while the effect is significantly negative for non-Spanish-speaking countries.[24] Finally, column (8) shows a significant 3.9 percent contraction in exports for countries that are both distant and non-Spanish-speaking, whereas column (7) reveals a negative, albeit not statistically significant effect for countries that are either close to Spain or Spanish-speaking.

Table 10 reports results from the same analysis but using exports to destinations other than Spain as the dependent variable. This serves as a placebo test: since the credit shock originated in Spain, we should not expect financial development or trade costs to play a significant role for other destinations, and indeed, we find no such effects. Taken together, these results confirm that financial

[23] Figure 12 plots the density of product-origin exposure by financial development group and shows that the distributions are similar, addressing concerns that our findings may be driven by a negative correlation between exposure and financial development.

[24] Figure 13 plots the density of product-origin exposure across country groups defined by trade costs and shows that the distributions are similar.

underdevelopment and trade costs, which arise from greater distance and language barriers, amplify spillovers specifically by constraining exporters' ability to attenuate the contraction in Spanish import demand.

One concern is that these estimates might partly reflect a reduction in lending by Spanish banks abroad, similar to the mechanism documented by Tripathy (2020). However, our results remain robust to the inclusion of exporter-year fixed effects, which already account for the presence of Spanish banks in foreign countries. That said, the trade channel could still be amplified by the Spanish banking footprint in foreign markets, for instance, if Spanish banks are specialized in financing firms that export to Spain (Paravisini et al., 2023).

To assess this mechanism, we use BankFocus data to map the international footprint of Spanish banks and compare export responses across countries with and without a Spanish bank presence. Reassuringly, the results in Table 11 show no differential response. Exports from financially developed countries remain unaffected regardless of Spanish bank presence, while the decline observed in less financially developed countries does not vary with local Spanish bank operations. Moreover, there is no evidence that Spanish bank presence interacts with trade costs. These findings rule out multinational banking networks as the primary driver of our results. Instead, the evidence supports a trade-based transmission mechanism, with financial underdevelopment and trade costs acting as key amplifying factors.

Additionally, we complement the previous results on the role of trade financing with firm-level evidence. In particular, we explore whether greater reliance on trade credit helps attenuate the negative effect on firm-level imports caused by the domestic bank policy.

Table 12 examines whether pre-policy reliance on supplier credit is associated with a smaller policy-induced decline in import growth.[25] Although our measure of supplier credit does not distinguish between domestic and foreign counterparties, Figure B.4 shows that firms more reliant on imports are also more likely to depend on supplier-provided credit, suggesting that foreign exporters are more likely to extend this facility. To account for this correlation, we include fixed effects by quintiles of import intensity interacted with time dummies, so that identification comes

[25] We measure reliance on supplier credit as the average share of supplier credit received by importers in total external financing (supplier credit plus bank debt).

from comparisons among firms with similar import exposure over time rather than from cross-sectional differences in import intensity. As shown in column 3, firms with higher pre-existing supplier credit exhibit significantly smaller import contractions following the regulation.

We next examine whether this attenuation margin varies with (foreign) exporter-side trade frictions. Figure 14, Panel A, plots binned averages of supplier credit against measures of trade frictions, controlling for import intensity. Firms more exposed to geographically distant suppliers or to partners without a free trade agreement rely less on supplier credit; the same pattern holds for a composite index combining distance and the absence of an FTA. Panel B shows that this relationship is stronger among firms with high import intensity, suggesting that the pattern operates through the payment terms associated with imported inputs rather than reflecting general supplier characteristics.

Table 13 provides further evidence on the mitigating role of supplier credit as a function of trade frictions. The table presents the implied attenuation effects for importers sourcing from low- and high-friction markets. Specifically, the row labeled *Attenuation when dummy = 0* reports the marginal effect of supplier credit in low-friction markets, while *Attenuation when dummy = 1* reports the corresponding effect in high-friction markets, computed as linear combinations of coefficients. The estimates show that attenuation is positive and statistically significant in high-friction markets across all specifications. This implies that, among importers sourcing from distant partners or countries without a free trade agreement, higher pre-policy reliance on supplier credit is associated with significantly smaller import contractions following the regulation. By contrast, attenuation in low-friction markets is smaller in magnitude and generally not statistically significant. This pattern is consistent with Figure 2: when trade credit frictions are more severe, supplier credit becomes a more effective margin for smoothing the effects of a domestic tightening in bank financing.

6.2 Export Reallocation

If foreign exporters cannot fully attenuate the contraction in Spanish demand through trade credit, they may instead redirect sales toward alternative destinations. In the model, such reallocation is optimal only when the import decline is large enough to outweigh the costs of redirecting exports.

Whether this threshold is reached depends on the importance of Spain as a market and on the frictions exporters face to reallocate products across destinations.

To capture this, we note that the effect of our treatment measure, defined in equation (15), on total exports may be amplified or dampened by a product-origin pair's reliance on Spanish demand. For instance, a pair may appear highly exposed if its Spanish importers rely heavily on treated banks, but if Spain represents only a small share of its overall exports, the aggregate exposure is effectively low. In such cases, reallocation may be economically unfeasible as predicted by the fixed reallocation cost in our model: If Spain represents a small share, the costs of redirecting sales to new destinations outweigh the benefits. To account for this, we construct the following measure of exporter exposure to the regulation in Spain, which weights our product-origin exposure by the share of each pair's exports directed to Spain:

$$\text{Aggregate Exposure}_{go} = \frac{X_{go\,\text{Spain}}}{X_{go}} \times \text{Exposure}_{go}. \tag{18}$$

Where $\frac{X_{go:\text{Spain}}}{X_{go}}$ denotes Spain's share of country o's exports of product g prior to the policy.[26] Under this definition, a foreign exporter o of product g is highly exposed when Spanish importers of g rely heavily on treated banks and Spain represents a substantial share of o's exports of that product. We then re-estimate the difference-in-differences equation (17), using this new exposure measure as the independent variable and total exports to all destinations for each product-origin pair as the dependent variable:

$$\frac{X_{got} - \overline{X}_{go}}{(X_{got} + \overline{X}_{go})/2} = \rho \times \text{Aggregate Exposure}_{go} \times Post_t + \delta_{go} + \delta_{x(g)t} + \delta_{x(o)t} + u_{got}, \tag{19}$$

Thus, our parameter of interest, ρ, measures the cross-border effects of Spanish loan loss provisions and is identified by comparing total sales from highly exposed exporters to those from less exposed ones, before and after the regulation. If Spain's trade partners faced no frictions in reallocating sales away from Spain, our exposure measure would have no effect on foreign countries' total exports. Conversely, if reallocation frictions matter, we would observe a significant decline in

[26] Shares are calculated using 2006 data, consistent with the exposure measure in equation (15).

trade partners' total exports.

We report our results in Table 14. Column (1) shows that a one-standard deviation higher aggregate exposure is associated with a 1.6 percent decline in total exports after the policy. Although not statistically significant at conventional levels, this aggregate effect reflects a significant 3.4 percent contraction in exports to Spain and an insignificant response of exports to other destinations, as shown in columns (2) and (3). To explore heterogeneity, we split product–country-of-origin cells into two groups based on Spain's share in their total sales, such that each group accounts for 50 percent of Spain's total imports.[27] Columns (4)-(6) show that aggregate exposure has no statistically significant effect among exporters for whom Spain represents only a minor share of sales. By contrast, columns (7)-(9) indicate that when Spain is a major buyer, aggregate exposure leads to a 2.8 percent decline in total exports, driven almost entirely by a 4.5 percent contraction in sales to Spain, with no evidence of reallocation to other destinations.[28]

Figure 15 presents the event-study graph corresponding to the average effect reported in column (7) of Table 14. The decline in total exports from highly exposed product-origin pairs appears only after the policy shock, while exports from differently exposed pairs followed parallel trends before 2011, supporting our identifying assumption. These results indicate that the increase in loan loss provisions in Spain generated aggregate effects abroad through the import demand channel, as reallocation to alternative destinations was limited for the average exporter-product pair.

To assess the role of reallocation frictions discussed in our model, we conduct a heterogeneity analysis at the product level. We test the hypothesis that homogeneous products, such as commodities, can be reallocated across countries relatively easily, whereas heterogeneous products–where buyers demand differentiated varieties–pose greater difficulties for reallocation. Through the lens of the model, homogeneous goods would have near zero fixed and variable reallocation costs. Using BACI data, we compute an implicit price for each HS 6-digit product–country pair by dividing

[27] As a result of this partition, pairs with a high Spanish share are those where exports to Spain account for more than 6 percent of their total sales.

[28] Table B.10 in the Appendix conducts a similar analysis, estimating the impact of our aggregate exposure in countries that are geographically distant from Spain and do not share a common language. Columns (2), (5), and (8) show that exports to Spain decline sharply in these higher-trade-cost countries, consistent with our main results, which in turn drives the aggregate export response in columns (1), (4), and (7). By contrast, the response of exports to other destinations is unrelated to the trade costs of exporting to Spain.

total value by total weight, and then calculate the variance of these prices at the HS 2-digit level. Products with price dispersion in the bottom quartile are classified as homogeneous, while those above this cutoff are defined as heterogeneous.

We report our results in Table 15, focusing on countries that were less able to attenuate the import demand shock from Spain–those with low financial development, greater geographic distance, and no common language–where the need for reallocation is expected to be strongest. Columns (1)-(3) present results for products with greater price dispersion (i.e., more heterogeneous goods). In this case, total exports fall by 9%, driven entirely by a contraction in sales to Spain, which is not offset by sales to other destinations. By contrast, column (4) shows that when products are relatively homogeneous, total exports remain unaffected. Columns (5)-(6) confirm that although exports to Spain still decline, homogeneous products are reallocated to other destinations, fully compensating for the decline in Spanish imports.[29]

Overall, these findings confirm that the extent of cross-border transmission depends on two frictions: attenuation frictions, which bind in financially underdeveloped and high-trade-cost economies, and reallocation frictions, which bind for differentiated products.

7 Conclusions

This paper provides evidence that domestic banking regulations generate cross-border spillovers through international trade, with unintended consequences for the trade flows of foreign partners.

Our analysis focuses on the sharp and unexpected 2012 increase in loan-loss provisioning requirements for Spanish banks' exposures to the construction and real estate sectors, an intervention plausibly unrelated to global trade conditions. Using the CIR, confidential firm-level trade transactions, and bilateral trade data from BACI, we examine the effects of this policy along three dimensions: credit supply to firms engaged in international trade, firm-level import outcomes, and the aggregate exports of Spain's trade partners.

We document that banks more exposed to the regulation reduced credit supply to their bor-

[29] Figure 16 plots the density of aggregate exposure across product groups defined by price dispersion and shows that the distributions are similar.

rowers; importers were unable to fully substitute toward alternative lenders or funding sources, translating into a broad-based import decline. Moreover, less exposed importers did not compensate for this decline, indicating limited reallocation within Spain.

Turning to the international implications, we show that the resulting decline in Spanish import demand generated significant spillovers to Spain's trade partners. Consistent with the mechanisms emphasized in our conceptual framework, exporters adjusted along two margins. First, exporters could attenuate the shock by extending trade credit to constrained importers. We find that this attenuation margin depends on exporter-side financial conditions: exporters located in financially developed economies were able to absorb the shock, whereas exporters in less financially developed countries experienced significantly larger export declines. Moreover, these effects are amplified when trade costs are higher, in geographically distant or non–Spanish-speaking countries.

Second, exporters could respond by reallocating exports toward alternative destinations. However, we find limited evidence of systematic reallocation. Exporters were able to redirect sales primarily for relatively homogeneous products, whereas differentiated goods exhibited limited adjustment and experienced larger declines in total exports. These patterns highlight the role of product-level adjustment frictions in shaping the international transmission of domestic financial shocks.

Taken together, our findings show that the cross-border consequences of banking regulation extend beyond international capital flows or multinational bank networks. Domestic financial regulations can also propagate internationally through trade linkages, with the magnitude of spillovers shaped by exporters' financial development, trade costs, and product differentiation.

The evidence presented in this paper carries important policy implications. Domestic policymakers should account for externalities beyond their borders when designing prudential policy, and international coordination may therefore be critical—not only to safeguard financial stability, but also to minimize unintended disruptions to global trade.

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Figures and tables

Table 1: Summary statistics

	Obs.	Mean	sd	p25	p50	p75
Panel A: *Bank-firm credit data*						
Log of Credit	1,288,007	5.65	1.65	4.61	5.64	6.65
Long-term credit ratio	1,288,007	54.33	41.08	6.07	59.32	100.00
Collateralization rate	1,288,007	13.40	30.25	0.00	0.00	0.00
NPL ratio	1,288,007	8.07	26.87	0.00	0.00	0.00
Credit share	1,288,007	23.30	22.83	6.34	15.29	32.85
Termination dummy	137,800	0.18	0.38	0.00	0.00	0.00
Panel B: *Firm data*						
Log of Total Bank Credit	316,925	7.17	1.60	6.12	7.09	8.13
Panel C: *Bank data*						
Exposure	83	36.76	16.71	25.70	34.95	47.36
Size	83	14.69	2.38	12.84	14.13	16.63
Capital ratio	83	9.56	6.46	6.66	8.04	10.31
Liquidity ratio	83	5.70	5.73	1.83	3.49	8.12
ROA	83	0.08	1.07	0.11	0.27	0.45
NPL ratio	83	5.39	4.26	2.75	4.84	6.61
Rural bank dummy	83	0.47	0.50	0.00	0.00	1.00
% Lending to X or M	83	34.66	21.11	18.57	34.18	48.59
Local Govt. Credit/Assets	83	2.67	6.20	0.32	1.50	3.08
Panel D: *Balance of Payments data*						
Exports, mid-point growth	649,050	-0.89	1.30	-2.00	-2.00	0.39
Imports, mid-point growth	558,540	-0.84	1.28	-2.00	-1.93	0.39
Exports, Entry dummy	274,933	0.35	0.48	0.00	0.00	1.00
Imports, Entry dummy	214,098	0.35	0.48	0.00	0.00	1.00
Exports, Exit dummy	229,050	0.36	0.48	0.00	0.00	1.00
Imports, Exit dummy	221,390	0.33	0.47	0.00	0.00	1.00
Panel E: *BACI data*						
Imports, mid-point growth	3,141,210	-0.54	1.07	-1.90	-0.28	0.23
Exports, mid-point growth	24,945	-0.07	0.40	-0.17	-0.00	0.11

In Panel A, bank-firm controls are computed using end of 2011 credit information. In Panel C, bank variables are calculated using supervisory financial statements at the end of 2011. All variables are defined in Appendix C.

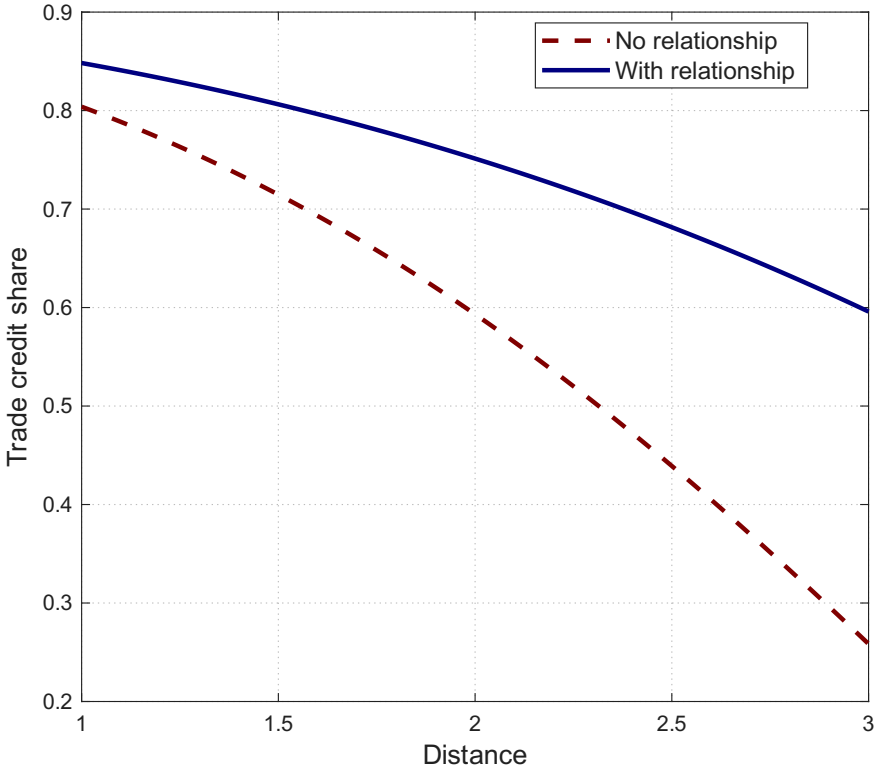
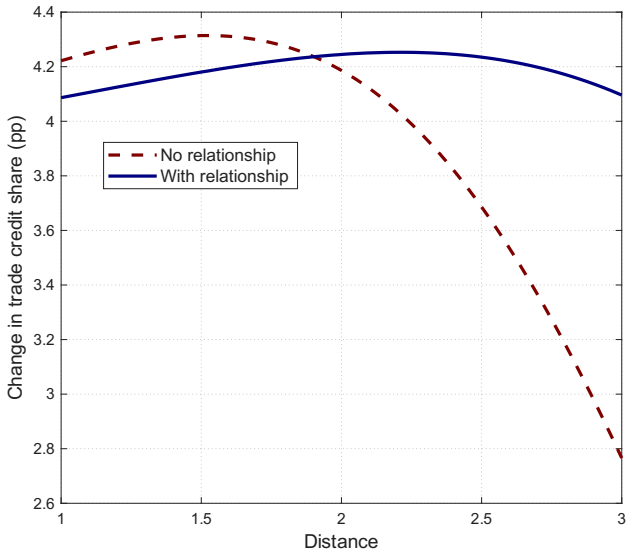
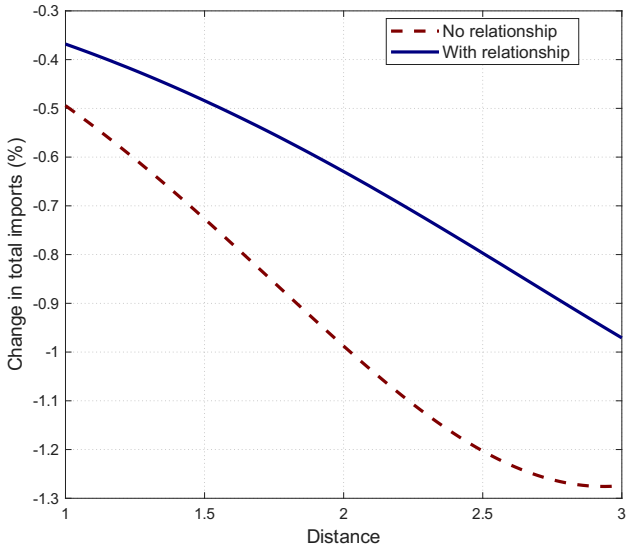


Figure 1: Trade Credit Usage by Distance

This figure plots the equilibrium share of imports financed through trade credit against country distance from the home market. The solid blue line corresponds to a scenario where importers have established lending relationships with suppliers prior to the shock, while the dashed red line depicts a scenario without such pre-existing relationships. While firms have access to trade credit in both cases, the recovery rate in default is lower in the absence of a prior relationship, particularly in distant markets.



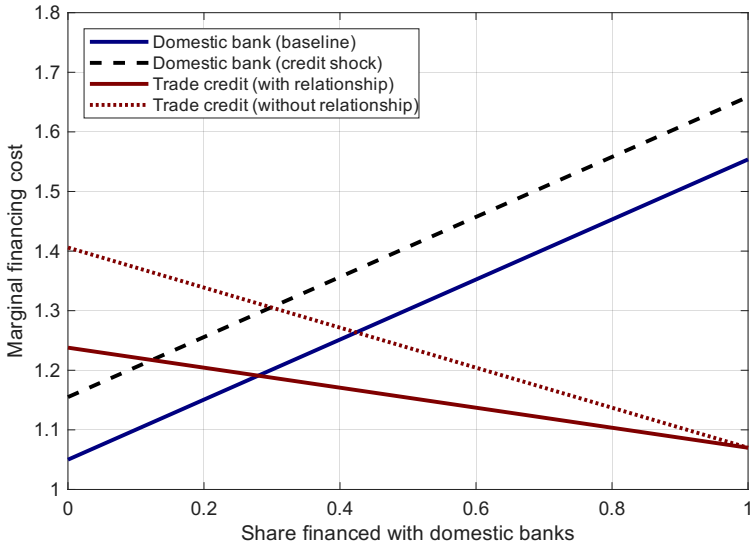
Panel A: Change in trade credit usage



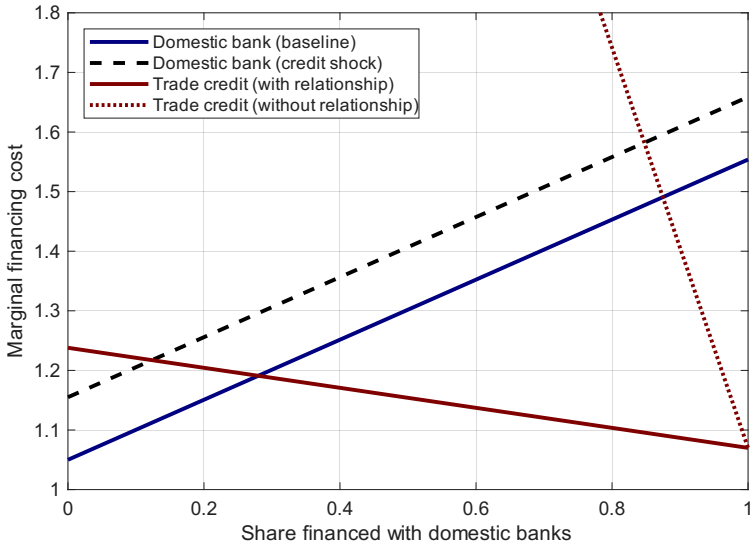
Panel B: Change in total imports

Figure 2: Response of Trade Credit Usage and Total Imports to a Domestic Credit Supply Shock

This figure illustrates how importers adjust their financing structure and total imports in response to a domestic credit supply shock, defined as an increase in $r_{b,0}$. Panel (a) shows the change in trade credit usage, measured in percentage points, as a function of country distance from the home market. Panel (b) reports the corresponding decline in total imports by distance. Solid lines represent importers with established supplier relationships; dashed lines represent those without.



Panel A: Close country



Panel B: Distant country

Figure 3: Marginal Cost of Domestic Bank Credit and Supplier Trade Credit

This figure presents a partial equilibrium analysis of an increase in $r_{b,0}$, holding total imports M fixed. Solid lines represent the marginal cost of domestic credit (blue) and trade credit (red) for importers with established supplier relationships; dashed red lines represent the marginal cost of trade credit for importers without such relationships. Panel (a) considers a close market and panel (b) considers a distant market. The marginal cost of domestic credit after the shock is the dashed dark line.

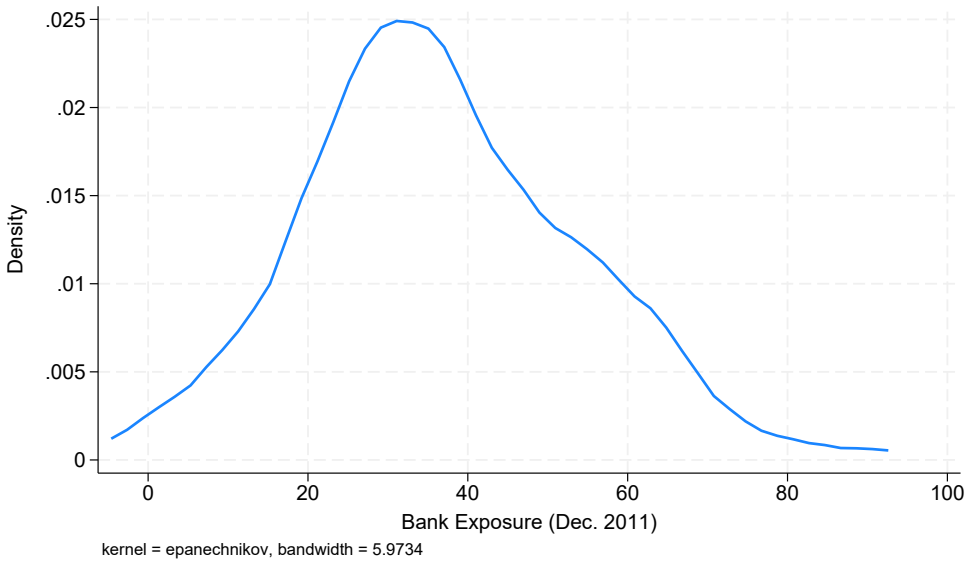


Figure 4: Bank Exposure: Kernel Density Estimate

This figure plots the kernel density of bank exposure defined by the ratio of construction and real estate lending over total corporate loans as of December 2011.

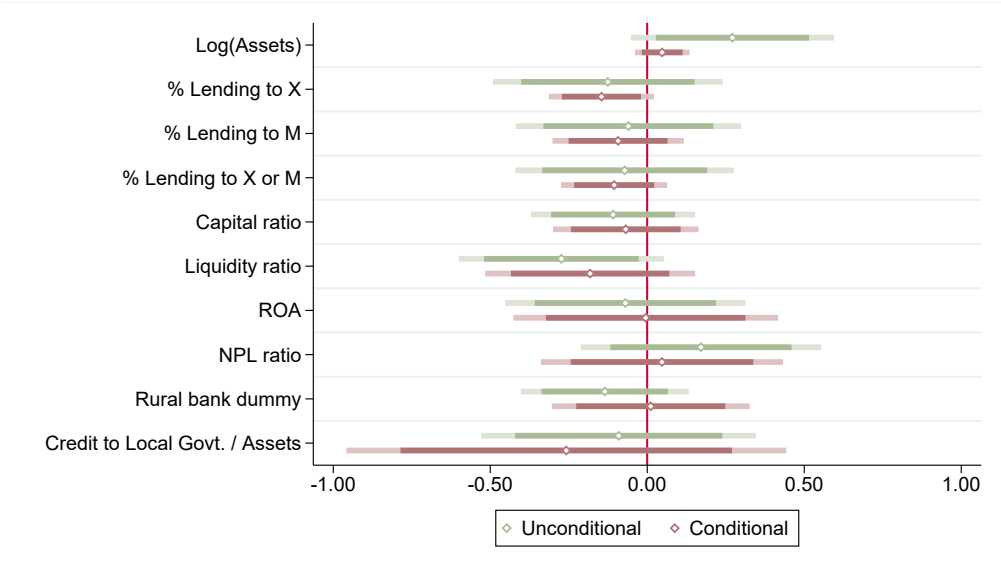


Figure 5: Bank Covariate Balance

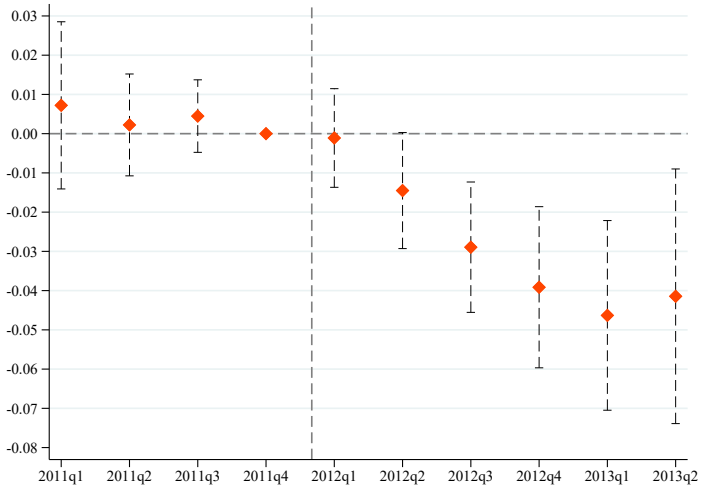
This figure shows coefficient estimates along with 95% confidence intervals (darker bars) and 99% confidence intervals (lighter bars) for the impact of a one standard deviation increase in bank exposure across different variables. All variables are standardized to have zero mean and a standard deviation of one. "Unconditional" estimates compare banks with different levels of exposure without conditioning on any fixed effects. "Conditional" estimates compare banks within the same size quartile (measured by total assets) and the same trade exposure quartile (measured by the share of loans to exporters and importers, excluding real estate and construction loans). Rural is an indicator variable equal to one for rural savings banks, and NPL represents non-performing loans, defined as outstanding debt with a repayment delay of more than 90 days.

Table 2: Impact of Provisioning Shift on Lending to Importing and Exporting Firms, Bank-Firm Level Analysis

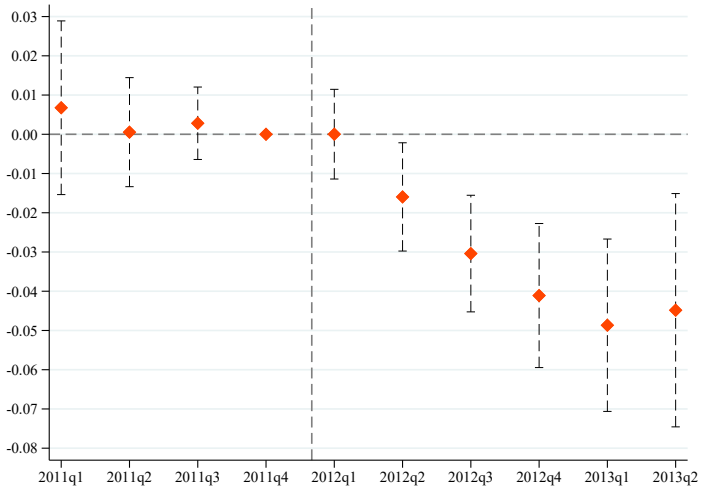
Panel A:	Intensive Margin			
Dep. Variable: Log(Credit)	*M*		*X*	
	(1)	(2)	(3)	(4)
Exposure × *Policy*	-0.025**	-0.031**	-0.025**	-0.031**
	(0.011)	(0.013)	(0.011)	(0.012)
Controls×time	No	Yes	No	Yes
Firm-time FE	Yes	Yes	Yes	Yes
Bank-firm FE	Yes	Yes	Yes	Yes
Obs.	962,963	962,963	973,386	973,386
R^2	0.97	0.97	0.97	0.97

Panel B:	Extensive Margin			
Dep. Variable: Termination	*M*		*X*	
	(1)	(2)	(3)	(4)
Exposure × *Policy*	0.016*	0.019*	0.014*	0.015
	(0.008)	(0.010)	(0.008)	(0.010)
Controls	No	Yes	No	Yes
Firm FE	Yes	Yes	Yes	Yes
Obs.	102,788	102,788	103,957	103,957
R^2	0.52	0.56	0.51	0.56

This table presents regression results on lending from bank b to firm f at the intensive margin (Panel A) and the extensive margin (Panel B). In Panel A, the dependent variable is the logarithm of credit granted by bank b to firm f in quarter t. In Panel B, conditional on the existence of the lending relationship at the end of 2011, the dependent variable is a dummy variable that takes the value of one if the lending relationship was terminated in the aftermath of the policy and zero otherwise. Columns (1) and (2) correspond to the sample of importers, whereas columns (3) and (4), exporters. *Exposure* is computed as the ratio of construction and real estate lending to total corporate sector lending by bank b as of the end of 2011. *Controls* are measured as of the end of 2011 and include bank-firm characteristics (the collateralization rate, the ratio of long-term lending, the share of firm f's NPLs with the bank, and the share of lending coming from bank b out of firm f's total bank lending) and bank characteristics (dummies for quartiles of size, measured by total assets, and quartiles of exposure to exporter/importer companies, measured as the lending share to exporters and importers relative to total corporate sector lending). All variables are defined in Appendix C. We exclude importing and exporting firms in the construction or real estate sectors. Standard errors are double clustered at the bank and firm levels and reported in parentheses. $***$ $p < 0.01$, $**$ $p < 0.05$, $*$ $p < 0.10$.



Panel A: Lending to Importers



Panel B: Lending to Exporters

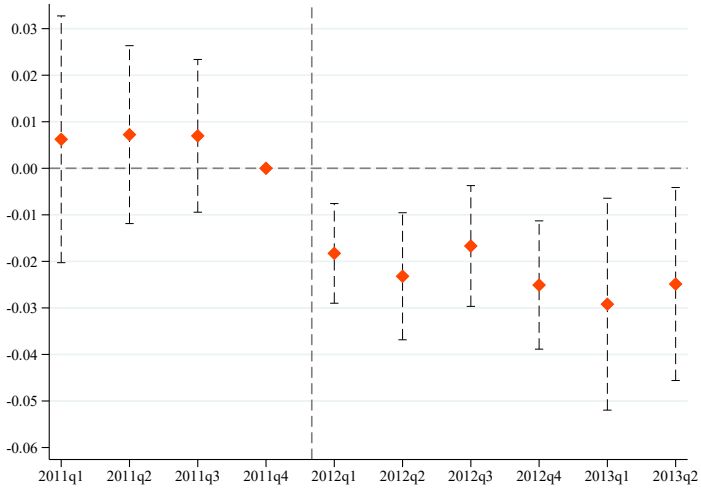
Figure 6: Impact on Lending to Importing and Exporting Firms at the Intensive Margin

This figure plots the event study associated with specification (12), where we replace *Policy* with quarter dummies. Panel (a) includes importers, while panel (b) consider exporters. The dependent variable is the logarithm of credit. We drop importing and exporting firms in construction or real estate sectors. Each dot represents the estimated coefficient of the interaction of bank exposure and a quarter dummy. We normalize the coefficient of 2011Q4 to zero. The dashed lines indicate the 2.5%–97.5% confidence interval, with standard errors double clustered at the bank and firm levels.

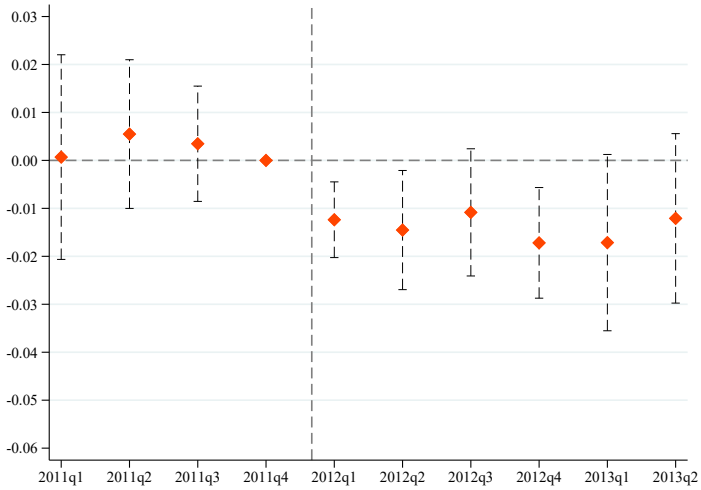
Table 3: Impact of Provisioning Shift on Total Bank Debt to Importing and Exporting Firms, Firm Level Analysis

Dep. Variable: Log(Credit)	M		X	
	(1)	(2)	(3)	(4)
Exposure × *Policy*	-0.019***	-0.028***	-0.012**	-0.016**
	(0.006)	(0.007)	(0.005)	(0.007)
Controls×time	No	Yes	No	Yes
Firm FE	Yes	Yes	Yes	Yes
Time FE	Yes	No	Yes	No
$\hat{\alpha}_{f,t}$	No	Yes	No	Yes
Obs.	228,786	228,786	230,099	230,099
R^2	0.98	0.98	0.98	0.98

This table presents firm-level regression results for firms' total bank debt, estimated using equation (13). Columns (1) and (2) correspond to the sample of importers, whereas columns (3) and (4), exporters. The dependent variable is the logarithm of firm f's total bank debt. *Exposure* is calculated as the weighted average of the exposure of banks lending to firm f, using outstanding credit at the end of 2011 as weights. To account for credit demand, regressions in columns (2) and (4) include estimated firm-quarter fixed effects obtained from specification (12), while regressions in columns (1) and (3) include quarter fixed effects instead. *Controls* are the weighted average of bank-firm characteristics (collateralization rate, ratio of long-term lending, share of firm f's NPLs with the bank) and bank characteristics (quartiles of size and exposure to exporter/importer companies), measured as of the end of 2011 and weighted by banks' lending to firm f as of 2011. Additionally, we include industry and province indicators as control variables. All variables are defined in Appendix C. Importing and exporting firms in the construction or real estate sectors are excluded. Standard errors are double-clustered at the main bank and firm levels and are reported in parentheses.
$***\ p < 0.01$, $**\ p < 0.05$, $*\ p < 0.10$.



Panel A: Lending to Importers



Panel B: Lending to Exporters

Figure 7: Impact on Total Bank Debt to Importing and Exporting Firms

This figure plots the event study associated with specification (13), where *Policy* is replaced with quarter dummies. Panel (a) includes importers, while panel (b) consider exporters. The dependent variable is the logarithm of firms' total bank credit. To account for credit demand, all regressions include the estimated firm-quarter fixed effects obtained from our bank-firm specification in (12). Importers and exporters in the construction or real estate sectors are excluded. Each dot represents the estimated coefficient of the interaction between a quarter dummy and the firm-level exposure, which is computed as the weighted average of the exposure of banks lending to firm f, using the outstanding credit at the end of 2011 as weights. The coefficient for 2011Q4 is normalized to zero. The dashed lines indicate the 2.5%–97.5% confidence interval, with standard errors double-clustered at the main bank level and firm level.

Table 4: Effect of Shift in Provisions on annual Firm-level Outcomes

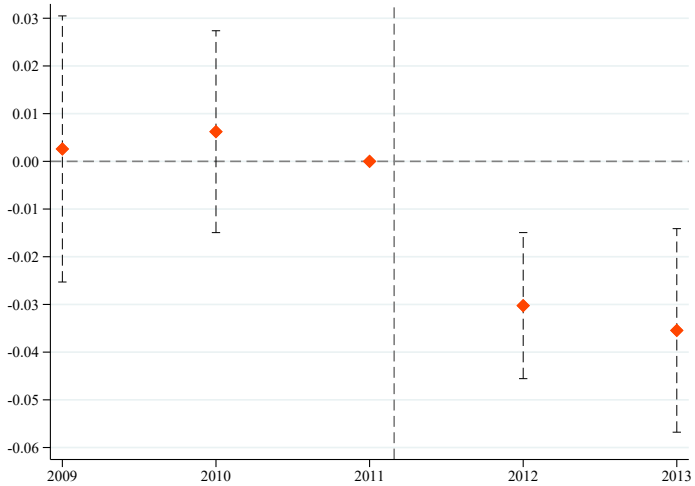
Panel A:	Importers				
	Bank Debt	Total Debt	Total Debt +Equity	Employment	Tangible Assets
	(1)	(2)	(3)	(4)	(5)
Exposure	-0.036***	-0.023*	-0.015*	-0.008*	-0.014**
	(0.013)	(0.013)	(0.009)	(0.004)	(0.005)
Firm controls	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes
Main Buyer FE	Yes	Yes	Yes	Yes	Yes
Obs.	17,269	17,269	17,269	17,269	17,269
R^2	0.02	0.02	0.02	0.02	0.02
Panel B:	Exporters				
	Bank Debt	Total Debt	Total Debt +Equity	Employment	Tangible Assets
	(1)	(2)	(3)	(4)	(5)
Exposure	-0.022	-0.009	0.036	-0.006	-0.008
	(0.014)	(0.012)	(0.047)	(0.004)	(0.006)
Firm controls	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes
Main Buyer FE	Yes	Yes	Yes	Yes	Yes
Obs.	17,194	17,194	17,194	17,194	17,194
R^2	0.01	0.02	0.02	0.02	0.03

This table reports regression results on firm-level outcomes using annual financial statements. The dependent variables are the mid-point growth rates between 2011 and 2012 of total bank debt, total debt, total debt plus equity, employment, and tangible assets. The main explanatory variable, *Exposure*, is computed as the weighted average exposure of banks lending to firm f, using outstanding credit at the end of 2011 as weights. Controls (measured in 2011) include weighted averages of bank characteristics—total assets and the share of lending to importing and exporting firms—as well as firm characteristics (log total assets, cash-to-assets, sales-to-assets, and own funds-to-assets ratios). Regressions include the fixed effects listed in the table, accounting for industry, province, and the firm's main export destination. All variables are defined in Appendix C. Standard errors are double-clustered at the main bank and firm levels. $***\ p < 0.01$, $**\ p < 0.05$, $*\ p < 0.10$.

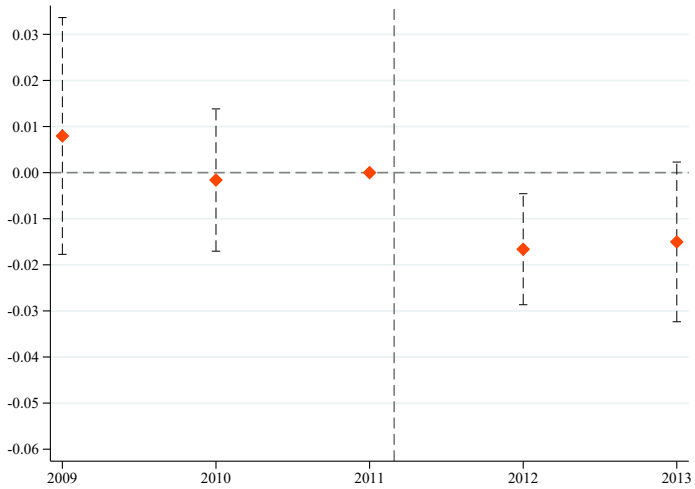
Table 5: Impact of Provisioning Shift on Firms' International Trade Flows

	Mid-point growth		Entry Dummy		Exit Dummy	
	M	X	M	X	M	X
	(1)	(2)	(3)	(4)	(5)	(6)
Exposure × *Policy*	-0.036***	-0.020**	-0.007**	0.000	0.014***	0.009**
	(0.011)	(0.009)	(0.003)	(0.003)	(0.004)	(0.004)
Product-Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Province-Industry-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Product-Country-Firm FE	Yes	Yes	No	No	No	No
Firm FE	No	No	Yes	Yes	Yes	Yes
Obs.	558,540	649,050	214,098	274,933	221,390	229,050
R^2	0.17	0.17	0.18	0.16	0.23	0.25

This table presents the regression estimates on firm's trade flows at the product-country-year level. In columns 1 and 2, we compute the mid-point growth rate between firm f's imports (exports) of product g to (from) country c and the 5-year average of firm f's imports (exports) of the same product from (to) that country, and regress it on the firm's exposure and a set of fixed effects. In columns 3 and 4, we employ a binary indicator that, conditional on not participating in the market the previous year, takes the value of one if the firm enters the market. In columns 5 and 6, we use a binary indicator that takes the value of one if the firm exits the market, conditional on participating in the same market the previous year. *Exposure* is computed as the average exposure of banks lending to firm f, with weights based on the outstanding credit as of the end of 2011. *Policy* is a dummy variable that takes the value of one after the provisioning increase. The fixed effects included in each regression are noted in the lower part of the table. All variables are defined in Appendix C. Importers and exporters in the construction or real estate sectors are excluded. Standard errors are double-clustered at the main bank and firm levels and are reported in parentheses. $***\ p < 0.01$, $**\ p < 0.05$, $*\ p < 0.10$.



(a) Imports



(b) Exports

Figure 8: Impact of Provisioning Shift on Imports and Exports at the Intensive Margin

This figure plots coefficient estimates from a modified version of the specification in (14). For each year, the coefficient corresponds to the interaction of *Exposure* with a year dummy. Each coefficient measures the impact of a one standard deviation increase in firm f's exposure to the policy on its imports (or exports) growth of good g from (or to) country c, relative to the year before the policy implementation (2011). Panel (a) presents the estimates for importers, whereas Panel (b) presents the estimates for exporters. Importers and exporters in the construction or real estate sectors are excluded. The dashed lines indicate the 2.5%–97.5% confidence interval, with standard errors double clustered at the main bank and firm levels.

Table 6: Bank-Firm Relationship Characteristics and Trade Flows

Panel A:	Importers				
	(1)	(2)	(3)	(4)	(5)
Exposure×Policy	-0.036***	-0.034***	-0.029**	-0.063***	-0.050***
	(0.011)	(0.011)	(0.011)	(0.010)	(0.010)
×Available funds in CLs		0.010*			0.012**
		(0.005)			(0.005)
×NPL			-0.134***		-0.111**
			(0.049)		(0.047)
×Share Main Bank				0.036***	0.028***
				(0.007)	(0.007)
Product-Country-Firm FE	Yes	Yes	Yes	Yes	Yes
Product-Country-Year FE	Yes	Yes	Yes	Yes	Yes
Province-Industry-Year FE	Yes	Yes	Yes	Yes	Yes
Observations	558,540	558,540	558,540	558,540	558,540
R^2	0.17	0.17	0.17	0.17	0.17
Panel B:	Exporters				
	(1)	(2)	(3)	(4)	(5)
Exposure × *Policy*	-0.020**	-0.019**	-0.010	-0.043***	-0.026***
	(0.009)	(0.009)	(0.009)	(0.010)	(0.010)
×Available funds in CLs		0.006			0.009
		(0.007)			(0.007)
×NPL			-0.093*		-0.072
			(0.048)		(0.047)
×Share Main Bank				0.033***	0.021***
				(0.005)	(0.006)
Product-Country-Firm FE	Yes	Yes	Yes	Yes	Yes
Product-Country-Year FE	Yes	Yes	Yes	Yes	Yes
Province-Industry-Year FE	Yes	Yes	Yes	Yes	Yes
Observations	649,050	649,050	649,050	649,050	649,050
R^2	0.17	0.17	0.17	0.17	0.17

This table presents the heterogeneous effects of the provisioning shift on trade outcomes based on lending relationship characteristics. The dependent variable is the mid-point growth rate of firm f's exports of product g relative to the 5-year average of firm f's exports of the same product. We regress this variable on the firm's exposure and a set of fixed effects. *Exposure* is computed as the weighted average of the exposure of banks lending to firm f, using outstanding credit as of the end of 2011 as weights. *Policy* takes a value of one for 2012 and 2013 and zero otherwise. *Available Funds in CLs* represents the share of undrawn to total committed funds available in firm f's credit lines as of the end of 2011. *Share Main Bank* is the share of firm f's outstanding credit held with its main bank as of the end of 2011. *NPL* takes a value of one if the firm had at least one non-performing loan as of the end of 2011 and zero otherwise. To facilitate the interpretation of the coefficients, *Available Funds in CLs* and *Share Main Bank* are standardized. The fixed effects included in each regression are specified in the lower part of the table. All variables are defined in Appendix C. Importers and exporters in the construction or real estate sectors are excluded. Standard errors are double-clustered at the main bank and firm levels and are reported in parentheses.
$***\ p < 0.01$, $**\ p < 0.05$, $*\ p < 0.10$.

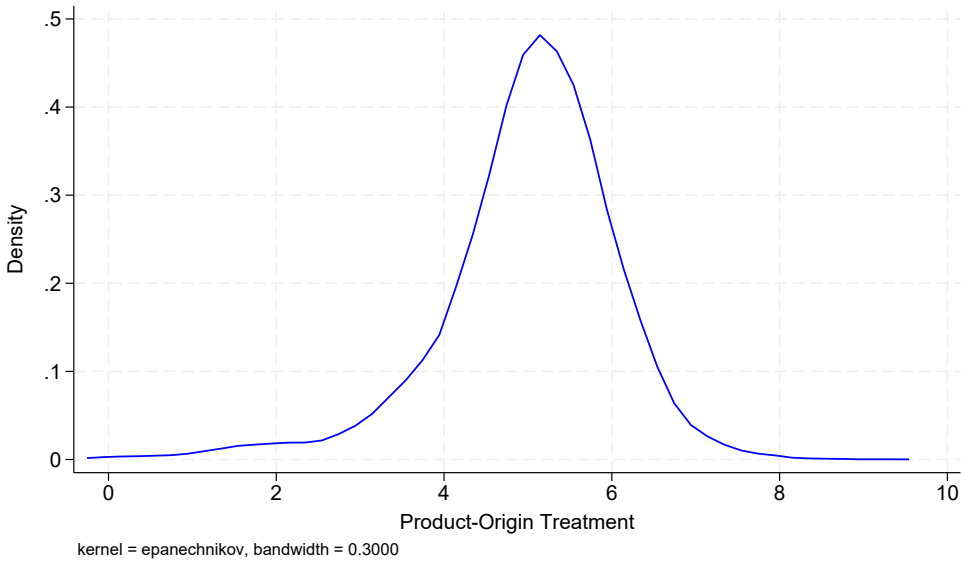


Figure 9: Product-origin Exposure: Kernel Density Estimate

This figure plots the kernel density of the standardized product-origin treatment, as defined in Equation 15.

Table 7: Impact of Provisioning Shift on Exports to Spain Relative to Other Destinations, Triple Difference-in-Differences Specification

	Mid-point growth				
	All destinations			Comparable destinations	All dest. Excl. GIIP exporters
	(1)	(2)	(3)	(4)	(5)
$Exposure_{go} \times \mathbb{1}[d = \text{Spain}] \times Post_t$	-0.032**	-0.027**	-0.021*	-0.022*	-0.023*
	(0.013)	(0.013)	(0.012)	(0.013)	(0.013)
Observations	1,600,910	1,600,910	1,600,910	144,500	1,394,170
Fixed Effects					
Exporter × Importer × Product	Yes	Yes	Yes	Yes	Yes
Importer × Year	Yes	Yes	Yes	Yes	Yes
Product × Year	No	Yes	Yes	Yes	Yes
Exporter × Year	No	No	Yes	No	Yes

This table presents the regression results for the mid-point growth rate of exports of product g from country o to country d in year t relative to its five-year average. The analysis is based on a product-origin level measure of exposure, interacted with a dummy variable that equals one when Spain is the importing country, and incorporates different sets of fixed effects, as specified in equation (16). Product-origin exposure is calculated as the weighted average of the exposures of Spanish banks lending to firms importing product g from country o, following equation (15). The fixed effects included in each regression are listed in the lower part of the table. Columns (1) to (3) include all importers, while column (4) focuses on a set of comparable destinations, as those in Autor et al. (2013). Column (5) includes all importers and excludes GIIP exporters. All variables and comparable countries are defined in Appendix C. Standard errors shown in parentheses are clustered by product and country of origin.
$***\ p < 0.01$, $**\ p < 0.05$, $*\ p < 0.10$.

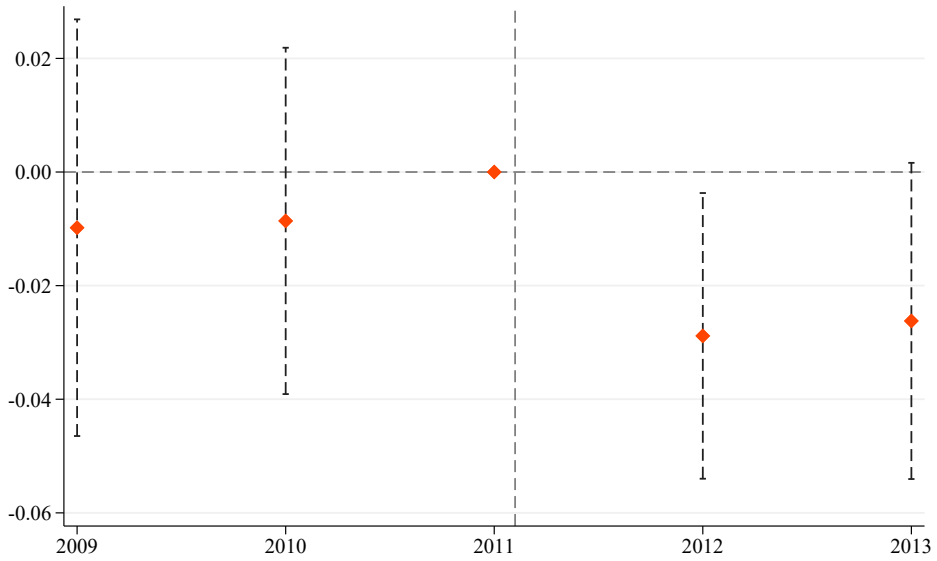


Figure 10: Event Study for the Average Effect of Product Exposure to Provisioning Shift on Exports to Spain Relative to Other Destinations

This figure plots coefficient estimates from a modified version of the specification in equation (16), which coincides with our benchmark specification in column (3) of Table 7. For each year, the coefficient corresponds to the interaction of the product-level exposure, the year dummy, and an indicator equal to one when Spain is the importing country. The dashed lines indicate the 2.5%–97.5% confidence interval, with standard errors clustered by product and country of origin.

Table 8: Impact of Provisioning Shift on Exports to Spain and Other Destinations

	Mid-point growth							
	Exports to Spain				Other Destinations			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$Exposure_{go} \times Post_t$	-0.034**	-0.035***	-0.027**	-0.028*	-0.003	-0.004	-0.010	-0.010
	(0.013)	(0.013)	(0.014)	(0.015)	(0.007)	(0.007)	(0.007)	(0.008)
Observations	11,363	11,363	11,363	9,948	11,363	11,363	11,363	9,948
Fixed Effects								
Exporter × Product	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Spanish share decile × Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Product charact. × Year	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Country charact. × Year	No	No	Yes	Yes	No	No	Yes	Yes

This table presents regression results for the mid-point growth rate of exports of product g from country o relative to its five-year average: to Spain (columns 1-3) and to destinations other than Spain (columns 5-7). Columns (4) and (8) report results excluding GIIP exporters. The analysis is based on a product-origin level measure of exposure, and incorporates different sets of fixed effects, as specified in equation (17). Product-origin exposure is calculated as the weighted average of the exposures of Spanish banks lending to firms importing product g from country o, following equation (15). The fixed effects included in each regression are listed in the lower part of the table. Product characteristics include a dummy for high or low heterogeneity: i.e, low, if price dispersion of the product is at the bottom quartile of the price dispersion distribution. Exporter country characteristics include: distance to Spain indicator (close to Spain if it belongs to the bottom quartile of the distance-to-Spain distribution), official language (Spanish or not). Spanish share corresponds to the participation of Spain in total exports of a specific origin×product pair. All variables are defined in Appendix C. Standard errors shown in parentheses are clustered by product and country of origin.
$***\ p < 0.01$, $**\ p < 0.05$, $*\ p < 0.10$.

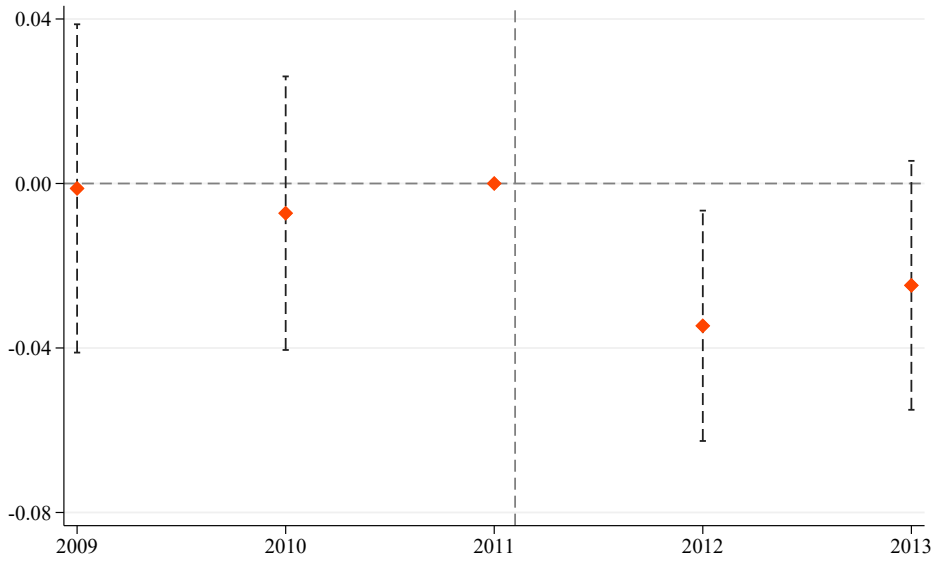


Figure 11: Event Study for the Average Effect of Product Exposure to Provisioning Shift on Exports to Spain

This figure plots coefficient estimates from a modified version of the specification in equation (17), which coincides with our benchmark specification in column (3) of Table 8. For each year, the coefficient corresponds to the interaction of the product-level exposure and the year dummy. The dashed lines indicate the 2.5%–97.5% confidence interval, with standard errors clustered by product and country of origin.

Table 9: Impact of Provisioning Shift on Exports to Spain, by Origin-Country Characteristics

	Mid-point growth Exports to Spain							
	High FD	Low Financial Development						
		All	Distance		Language		Far & Not Spanish	
			Close	Far	Spanish	Other	No	Yes
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$Exposure_{go} \times Post_t$	0.005	-0.033**	-0.022	-0.036**	-0.026	-0.035**	-0.023	-0.039**
	(0.024)	(0.016)	(0.034)	(0.018)	(0.055)	(0.017)	(0.031)	(0.019)
Observations	2,703	8,271	1,620	6,651	1,040	7,231	2,660	5,611
Fixed Effects								
Exporter × Product	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Product charact. × Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country charact. × Year	Yes	Yes	Yes	Yes	Yes	Yes	–	–
Spanish share decile × Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

This table presents the regression results for the mid-point growth rate of exports of product g from country o to Spain, relative to its five-year average. Column (1) reports estimates for exports to Spain from countries with high financial development, defined as those in the top quartile of the private credit-to-GDP ratio. Columns (2)-(8) report estimates for countries with low financial development, those at the bottom three quartiles. Column (2) includes all origins. Columns (3) and (4) correspond to exporters that are geographically close to Spain (bottom quartile of the distance-to-Spain distribution), or far from Spain, respectively. Columns (5) and (6) report estimates for exporters from Spanish-speaking and non-Spanish-speaking countries, respectively. Column (7) includes exporters that are either geographically close to Spain or Spanish-speaking, while Column (8) covers exporters that are both geographically distant and non-Spanish-speaking. The analysis is based on a product-origin level measure of exposure, and incorporates different sets of fixed effects, as specified in equation (17). Product-origin exposure is calculated as the weighted average of the exposures of Spanish banks lending to firms importing product g from country o, following equation (15). The fixed effects included in each regression are listed in the lower part of the table. Product characteristics include a dummy for high or low heterogeneity: i.e, low, if price dispersion of the product is at the bottom quartile of the price dispersion distribution. Exporter country characteristics include: distance to Spain indicator (close to Spain or not), official language (Spanish or non Spanish). Spanish share corresponds to the participation of Spain in total exports of a specific origin×product pair. All variables are defined in Appendix C. Standard errors shown in parentheses are clustered by product and country of origin.
$***\ p < 0.01$, $**\ p < 0.05$, $*\ p < 0.10$.

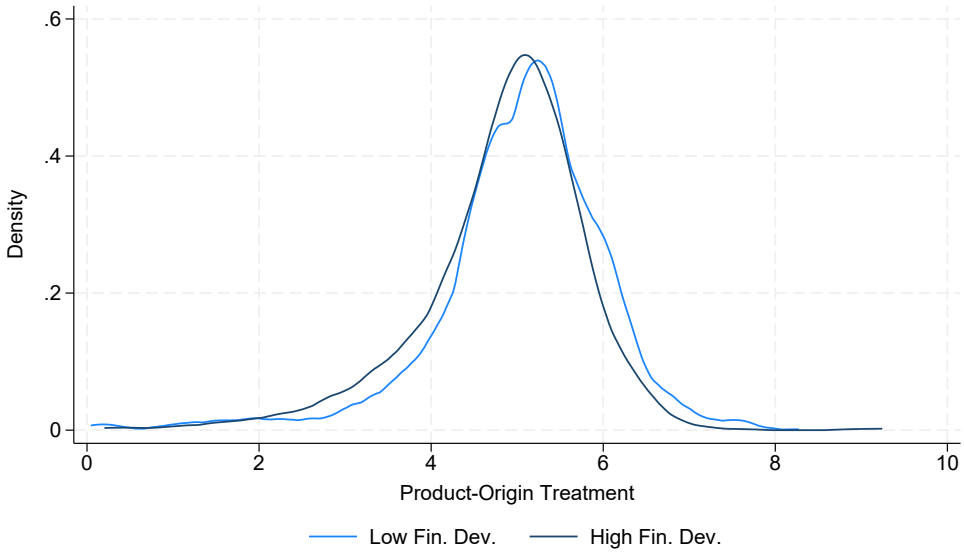


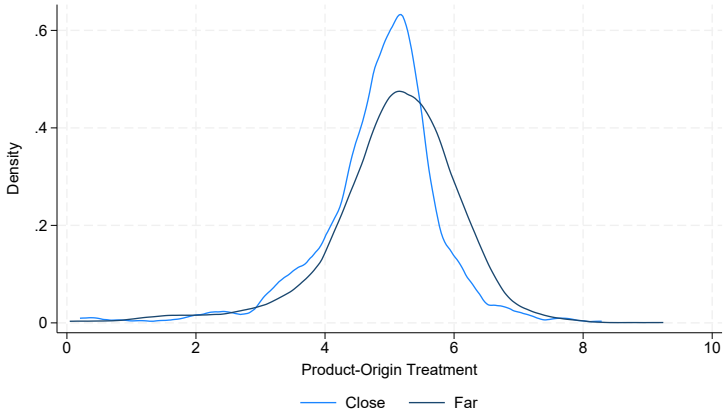
Figure 12: Product-origin Exposure: Kernel Density Estimates by Financial Development of Origin Country

This figure plots the kernel density of the standardized product-origin treatment, as defined in Equation 15, by the level of financial development in the origin country, where financially developed countries are defined as those in the top quartile of the private credit-to-GDP ratio.

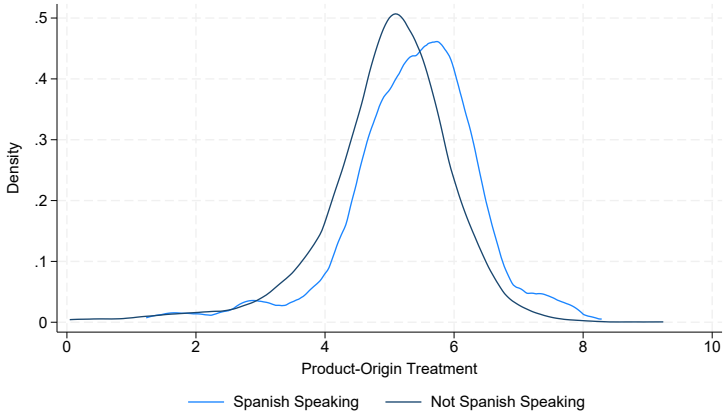
Table 10: Impact of Provisioning Shift on Exports to Destinations other than Spain, by Origin-Country Characteristics

	Mid-point growth Exports to Destinations other than Spain							
	High FD	Low Financial Development						
		All	Distance		Language		Far & Not Spanish	
			Close	Far	Spanish	Other	No	Yes
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$Exposure_{go} \times Post_t$	0.001	-0.008	-0.015	-0.008	-0.034	-0.005	-0.022	-0.004
	(0.013)	(0.008)	(0.020)	(0.009)	(0.026)	(0.008)	(0.015)	(0.009)
Observations	2,703	8,271	1,620	6,651	1,040	7,231	2,660	5,611
Fixed Effects								
Exporter × Product	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Product charact. × Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country charact. × Year	Yes	Yes	Yes	Yes	Yes	Yes	–	–
Spanish share decile × Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

This table presents the regression results for the mid-point growth rate of exports of product g from country o to destinations other than Spain, relative to its five-year average. Column (1) reports estimates for exports to Spain from countries with high financial development, defined as those in the top quartile of the private credit-to-GDP ratio. Columns (2)-(8) report estimates for countries with low financial development, those at the bottom three quartiles. Column (2) includes all origins. Columns (3) and (4) correspond to exporters that are geographically close to Spain (bottom quartile of the distance-to-Spain distribution), or far from Spain, respectively. Columns (5) and (6) report estimates for exporters from Spanish-speaking and non-Spanish-speaking countries, respectively. Column (7) includes exporters that are either geographically close to Spain or Spanish-speaking, while Column (8) covers exporters that are both geographically distant and non-Spanish-speaking. The analysis is based on a product-origin level measure of exposure, and incorporates different sets of fixed effects, as specified in equation (17). Product-origin exposure is calculated as the weighted average of the exposures of Spanish banks lending to firms importing product g from country o, following equation (15). The fixed effects included in each regression are listed in the lower part of the table. Product characteristics include a dummy for high or low heterogeneity: i.e, low, if price dispersion of the product is at the bottom quartile of the price dispersion distribution. Exporter country characteristics include: distance to Spain indicator (close to Spain or not), official language (Spanish or non Spanish). Spanish share corresponds to the participation of Spain in total exports of a specific origin×product pair. All variables are defined in Appendix C. Standard errors shown in parentheses are clustered by product and country of origin.
$***\ p < 0.01$, $**\ p < 0.05$, $*\ p < 0.10$.



(a) Distance



(b) Language

Figure 13: Product-origin Exposure: Kernel Density Estimates by Distance and Language

The figure plots the kernel density of the standardized product-origin treatment, as in Equation 15. Panel A splits origin countries into those geographically close to Spain–defined as the bottom quartile of the distance-to-Spain distribution–and those farther away. Panel B divides origin countries into Spanish-speaking and non-Spanish-speaking.

Table 11: Impact of Provisioning Shift on Exports to Spain by Presence of Spanish Banks

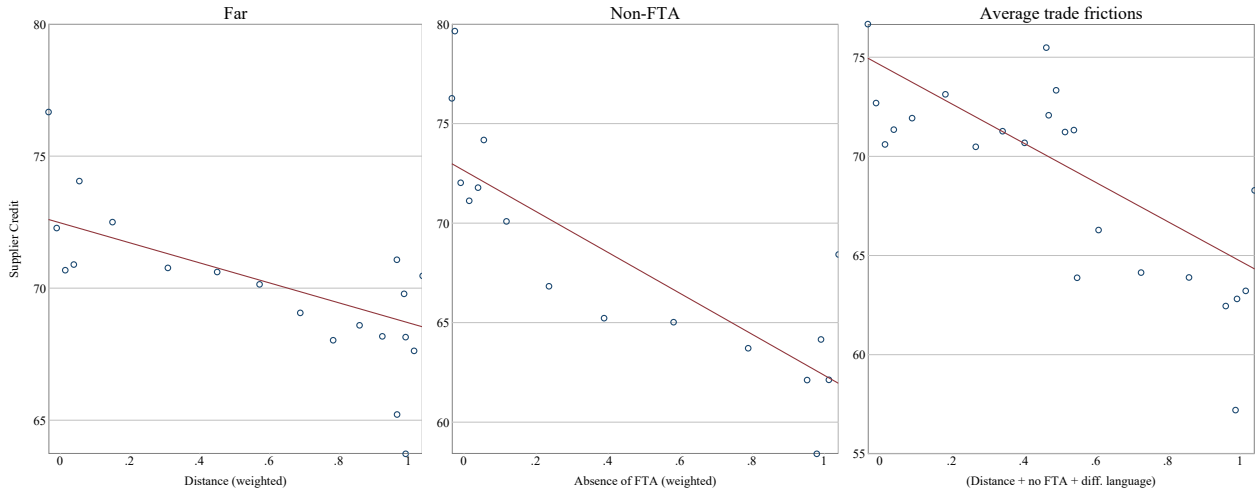
	Mid-point growth Exports to Spain							
	High FD		Low Financial Development					
					Far		Not Spanish	
	Without	With	Without	With	Without	With	Without	With
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$Exposure_{go} \times Post_t$	0.004	-0.016	-0.037**	-0.032	-0.039*	-0.037	-0.034*	-0.057*
	(0.032)	(0.028)	(0.018)	(0.032)	(0.020)	(0.037)	(0.018)	(0.032)
Observations	1,788	915	6,571	1,700	5,366	1,285	6,196	1,035
Fixed Effects								
Exporter × Product	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Product charact. × Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country charact. × Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Spanish share decile × Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

This table presents the regression results for the mid-point growth rate of exports of product g from country o to Spain, relative to its five-year average. Columns (1)-(2) reports estimates for exports to Spain from countries with high financial development, defined as those in the top quartile of the private credit-to-GDP ratio. Columns (3)-(8) report estimates for countries with low financial development, those at the bottom three quartiles. Columns (5)-(6) report estimates for countries that are geographically distant from Spain (top three quartiles of the distance-to-Spain distribution), columns (7)-(8) report estimates for countries that are not Spanish speakers. Columns (1), (3), (5), and (7) report estimates for exporting countries without a Spanish bank presence. Columns (2), (4), (6), and (8) report estimates for exporting countries where Spanish banks are present. The analysis is based on a product-origin level measure of exposure, and incorporates different sets of fixed effects, as specified in equation (17). Product-origin exposure is calculated as the weighted average of the exposures of Spanish banks lending to firms importing product g from country o, following equation (15). The fixed effects included in each regression are listed in the lower part of the table. Product characteristics include a dummy for high or low heterogeneity: i.e, low, if price dispersion of the product is at the bottom quartile of the price dispersion distribution. Exporter country characteristics include: distance to Spain indicator (close to Spain or not), official language (Spanish or non Spanish). Spanish share corresponds to the participation of Spain in total exports of a specific origin×product pair. All variables are defined in Appendix C. Standard errors shown in parentheses are clustered by product and country of origin.
$***\ p < 0.01$, $**\ p < 0.05$, $*\ p < 0.10$.

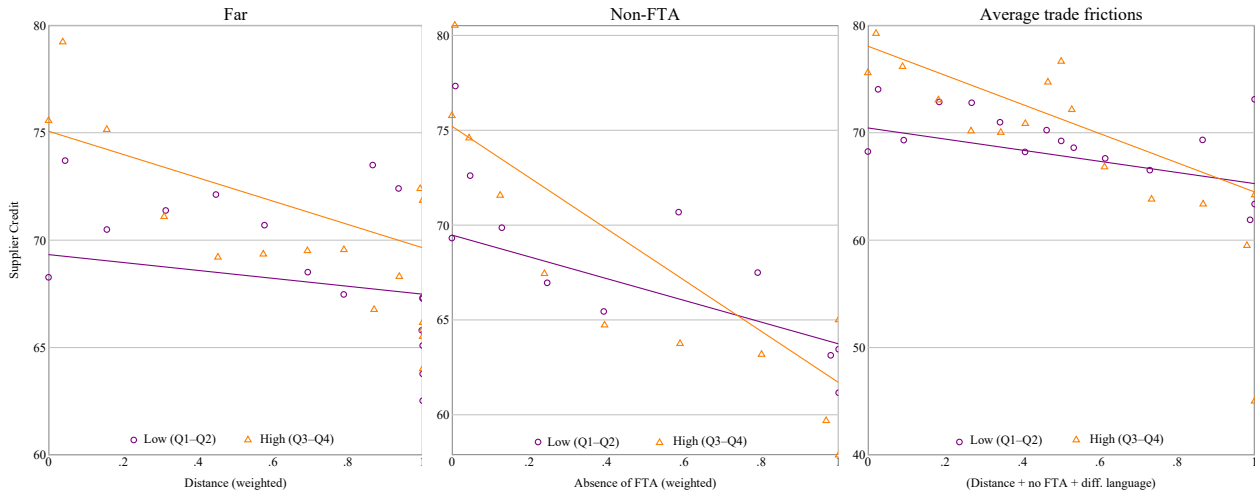
Table 12: Reliance on Trade Credit and Policy Impact on Trade Flows

	Importers		
	(1)	(2)	(3)
Exposure × *Policy*	-0.036***	-0.033***	-0.032**
	(0.011)	(0.011)	(0.013)
Exposure × *Policy* × *Supplier Credit*			0.020**
			(0.008)
Product-Country-Year FE	Yes	Yes	Yes
Province-Industry-Year FE	Yes	Yes	Yes
Product-Country-Firm FE	Yes	Yes	Yes
Imports/Purchases Bins-Year FE	No	No	Yes
Obs.	558,540	519,335	519,265
R^2	0.17	0.17	0.18

This table presents the heterogeneous effects of the provisioning shift on trade outcomes based on firms' reliance on trade credit, specifically, importer-received credit in Column (3). *Supplier Credit* is defined as the average share of supplier credit, received by importers, in total financing (supplier credit plus bank debt) over the same period. This measure is standardized to have mean zero and standard deviation one. Column (1) replicate the main specification from Table 5 using the full sample. Column (2) restricts the sample to firms for which trade credit data are available. Lower-order interaction terms corresponding to the triple interactions are included in the regressions but not reported. Column (3) additionally includes fixed effects at the year-imports share bins level. Bins are constructed using quintiles of the firm's average imports-to-purchases ratio over 2010-2011. The dependent variable is the mid-point growth rate of firm f's imports of product g relative to the 5-year average of firm f's imports of the same product. *Exposure* is computed as the weighted average of the exposure of banks lending to firm f, using outstanding credit as of the end of 2011 as weights. *Policy* takes a value of one for 2012 and 2013 and zero otherwise. The fixed effects included in each regression are specified in the lower part of the table. Importers operating in the construction or real estate sectors are excluded. Standard errors are double-clustered at the main bank and firm levels and are reported in parentheses. $***\ p < 0.01$, $**\ p < 0.05$, $*\ p < 0.10$.



(a) Supplier Credit and Trade Frictions (Controlling for Import Intensity)



(b) Supplier Credit and Trade Frictions by Import Intensity

Figure 14: Supplier Credit and Exporter Trade Frictions

This figure plots binned averages of supplier credit against firm-level exposure to trade frictions. Trade-friction measures are constructed as weighted averages, where weights correspond to the share of imports sourced from origins classified as geographically distant (Far) or without a free trade agreement (No FTA). We also report a composite index combining these dimensions. Panel (a) presents the relationship controlling for import intensity. Panel (b) reports the same relationships separately by import intensity; low import intensity corresponds to the bottom two quartiles of imports over total purchases. Supplier credit is defined as the average pre-policy share of supplier credit in total external financing (supplier credit plus bank debt).

Table 13: Trade Credit Attenuation and Exporter Trade Frictions

	(1)	(2)	(3)
	Far & No FTA (1/0)	Far (1/0)	No FTA (1/0)
Exposure × *Policy*	-0.029**	-0.034**	-0.028**
	(0.013)	(0.013)	(0.013)
Attenuation when dummy =0			
Exposure × *Policy* × *Supplier Credit*	0.015*	0.018	0.015
	(0.009)	(0.013)	(0.009)
Attenuation when dummy = 1			
Exposure × *Policy* × *Supplier Credit* +			
Exposure × *Policy* × *Supplier Credit* × *dummy*	0.034**	0.021**	0.034**
	(0.014)	(0.009)	(0.014)
Product-Country-Year FE	Yes	Yes	Yes
Province-Industry-Year FE	Yes	Yes	Yes
Product-Country-Firm FE	Yes	Yes	Yes
Imports/Purchases Bins-Year FE	Yes	Yes	Yes
Obs.	511,365	510,080	510,080
R^2	0.176	0.175	0.175

Notes: This table reports heterogeneous effects of trade credit in attenuating the impact of the provisioning policy shock on firm-level import growth, depending on exporter characteristics. The row labeled *Attenuation when dummy = 0* reports the coefficient on *Exposure* × *Policy* × *Supplier Credit*. The row labeled *Attenuation when dummy = 1* reports the sum of *Exposure* × *Policy* × *Supplier Credit* and its interaction with the corresponding dummy, computed using linear combinations of coefficients. The underlying quadruple interaction term *Exposure* × *Policy* × *Supplier Credit* × *Dummy* is included in all regressions but not reported. Standard errors for linear combinations are obtained using the delta method. Column (1) uses a dummy equal to one for exporters that are both geographically distant (Far) and do not share a free trade agreement with the importer (No FTA). Column (2) uses a dummy for distant exporters (Far), and column (3) uses a dummy for exporters without a free trade agreement (No FTA).*Supplier Credit* is defined as the average share of supplier credit received by importers in total external financing (supplier credit plus bank debt) and is standardized to have mean zero and unit standard deviation. Lower-order interaction terms are included in all regressions but not reported. Bins are constructed using quintiles of the firm's average imports-to-purchases ratio over 2010–2011. The dependent variable is the mid-point growth rate of firm f's imports of product g relative to its five-year average. *Policy* equals one in 2012–2013 and zero otherwise. Importers in construction and real estate are excluded. Standard errors are double-clustered at the main bank and firm levels and reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 14: Impact of Provisioning Shift on the Total Exports of Spain's Trade Partners, Conditional on Spain's share in their total sales

	Mid-point growth rate of Exports								
	All Product × Origin Pairs			Low Spanish Share			High Spanish Share		
	All	Spain	Rest	All	Spain	Rest	All	Spain	Rest
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$Agg.\ Exposure_{go} \times Post_t$	-0.016	-0.034**	0.007	-0.041	-0.082	-0.034	-0.028*	-0.045**	-0.002
	(0.011)	(0.015)	(0.012)	(0.075)	(0.119)	(0.076)	(0.015)	(0.018)	(0.015)
Observations	11,363	11,363	11,363	8,525	8,525	8,525	2,838	2,838	2,838
Fixed Effects									
Exporter × Product	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Product charact. × Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country charact. × Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

This table reports regression results for the mid-point growth rate of exports of product g from country o, relative to its five-year average. Columns (1), (4), and (7) present results for total exports to all destinations; columns (2), (5), and (8) for exports to Spain; and columns (3), (6), and (9) for exports to destinations other than Spain. Columns (1)-(3) include all product-origin pairs. Columns (4)-(6) restrict the sample to product-origin pairs with a low Spanish share, (i.e. Spanish purchases account for less than 6 percent of pair's total exports). Columns (7)-(9) restrict the sample to product-origin pairs with a high Spanish share. The analysis is based on an aggregate measure of exposure, and incorporates different sets of fixed effects, as specified in equation (19). Aggregate exposure is calculated as the product-origin level exposure defined in equation (15) multiplied by Spain's share of country o's exports of product g, following equation (18). The fixed effects included in each regression are listed in the lower part of the table. Product characteristics include a dummy for high or low heterogeneity: i.e, low, if price dispersion of the product is at the bottom quartile of the price dispersion distribution. Exporter country characteristics include: distance to Spain indicator (close to Spain if it belongs to the bottom quartile of the distance-to-Spain distribution), official language (Spanish or non Spanish). All variables are defined in Appendix C. Standard errors shown in parentheses are clustered by product and country of origin.
$***\ p < 0.01$, $**\ p < 0.05$, $*\ p < 0.10$.

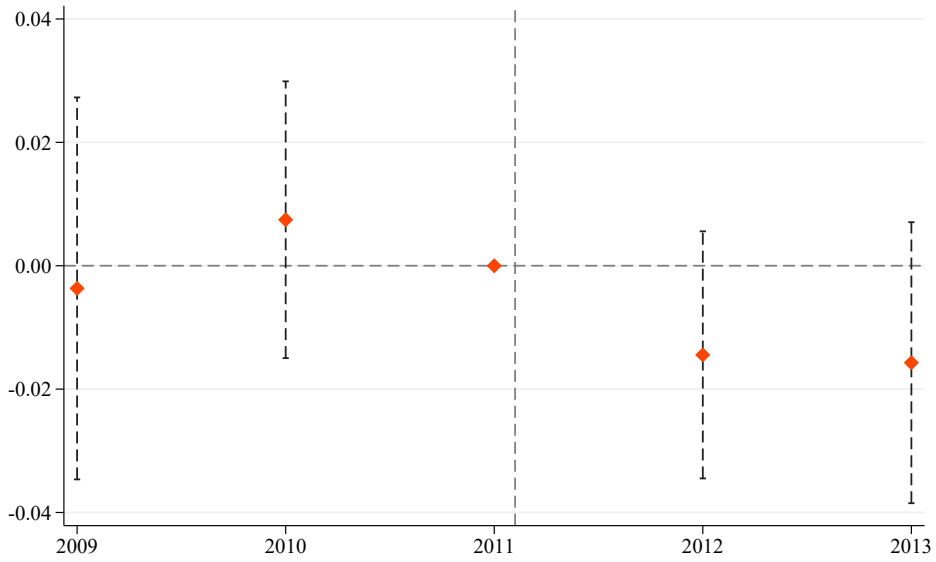


Figure 15: Event Study for the Average Effect of Aggregate Exposure to Provisioning Shift on Total Exports of Spain's Trade Partners, conditional on high Spanish share

This figure plots coefficient estimates from a modified version of the specification in equation (19), which coincides with our benchmark specification in column (7) of Table 14. For each year, the coefficient corresponds to the interaction of the aggregate exposure and the year dummy. The dashed lines indicate the 2.5%–97.5% confidence interval, with standard errors clustered by product and country of origin.

Table 15: Impact of Provisioning Shift on the Total Exports of Spain's Trade Partners and Product Differentiation: Low Financial Development and High Trade Costs Origins

	Mid-point growth rate of Exports					
	High Price Dispersion			Low Price Dispersion		
	All	Spain	Rest	All	Spain	Rest
	(1)	(2)	(3)	(4)	(5)	(6)
$Agg.\ Exposure_{go} \times Post_t$	-0.088**	-0.160***	-0.026	0.057	-0.291***	0.121**
	(0.037)	(0.049)	(0.025)	(0.052)	(0.098)	(0.059)
Observations	4,172	4,172	4,172	1,439	1,439	1,439
Fixed Effects						
Exporter × Product	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes

This table reports regression results for the mid-point growth rate of exports of product g from country o, relative to its five-year average. Columns (1) and (4) present results for total exports to all destinations; columns (2) and (5) for exports to Spain; and columns (3) and (6) for exports to destinations other than Spain. Columns (1)-(3) restrict the sample to products with high price dispersion, defined as those in the top three quartiles of the distribution. Columns (4)-(6) restrict the sample to products with low price dispersion (bottom quartile). The analysis is based on an aggregate measure of exposure, and incorporates different sets of fixed effects, as specified in equation (19). Aggregate exposure is calculated as the product-origin level exposure defined in equation (15) multiplied by Spain's share of country o's exports of product g, following equation (18). The fixed effects included in each regression are listed in the lower part of the table. All variables are defined in Appendix C. Standard errors shown in parentheses are clustered by product and country of origin.
$***\ p < 0.01$, $**\ p < 0.05$, $*\ p < 0.10$.

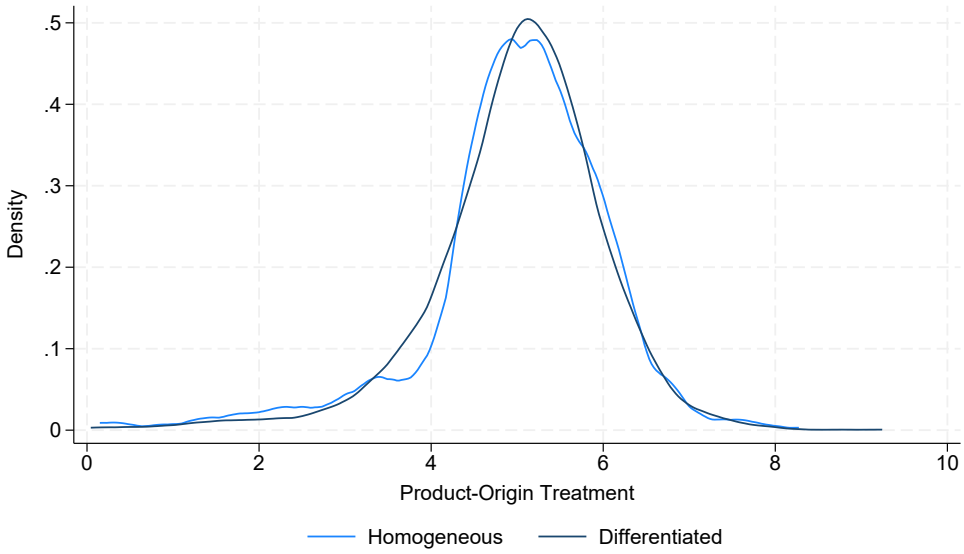


Figure 16: Product-origin Exposure: Kernel Density Estimates by Product Characteristics

This figure plots the kernel density of the standardized product-origin treatment, as in Equation 15, for heterogeneous products–defined as those in the top three quartiles of the distribution of HS6 price dispersion distribution–and homogeneous products.

A Theoretical framework proofs

Proposition 1. *The share of imports financed through foreign trade credit declines with distance between trading partners. For any given distance, importers with prior relationships rely more on trade credit than those without.*

Proof. In our setting, distance is negatively related to recovery rate α, while prior relationships are positively related. Thus, to prove this proposition, we need to show that $\partial s/\partial\alpha$ is positive, where $s = F/\tau M$. For simplification, we assume $C = 0$ and $z \sim \text{Exp}(\eta)$ along this proof.

We modify the problem as follows. Instead of assuming that the importer chooses F and L, we assume it chooses total imports M and the fraction $s \in [0, 1]$ of the financing need τM that is covered by trade credit. Note that the problem is exactly the same, but this formulation simplifies the analysis as we want to measure $\partial s/\partial\alpha$.

Default on trade credit occurs if productivity is below the threshold $T = \frac{r_b L + r_x F}{\ln M}$. Under our assumption of a exponential productivity function with parameter η importer's expected profit is:

$$\Pi(M, s) = \frac{\ln M}{\eta}\, e^{-\eta T} \tag{A.1}$$

In this formulation, domestic bank credit is $L = \tau M(1 - s)$ and foreign supplier credit is $F = \tau M s$. Thus, the threshold T can be expressed as follows:

$$T = \frac{\tau M\big[r_b(1 - s) + r_x s\big]}{\ln M} \tag{A.2}$$

Interest rate of domestic bank $r_b = r_{b,0} + \lambda_b L$ becomes:

$$r_b = r_{b,0} + \lambda_b \tau M(1 - s) \tag{A.3}$$

Trade credit interest rate under our zero-profit condition is $r_x = \frac{r_{x,0} - \chi\alpha}{1-\chi}$. The probability of default under the exponential productivity assumption is $\chi = 1 - e^{-\eta T}$. Thus, interest rate on trade credit can be expressed as follows:

$$r_x = \alpha + (r_{x,0} - \alpha)e^{\eta T} \tag{A.4}$$

The importer maximizes $\Pi(M, s)$ subject to equations (A.2), (A.3), and (A.4).

The first order condition respect to s leads to the following equation that equalizes marginal costs:

$$r_x = r_{b,0} + 2\lambda_b \tau M(1 - s)$$

This equation allows us to eliminate r_x from the definition of T. Thus, at the optimal financing mix, we have:

$$T(M, s) = \frac{\tau M\left[r_{b,0} + \lambda_b \tau M(1 - s^2)\right]}{\ln M} \tag{A.5}$$

From the definition of trade credit interest rate (A.4), our equalized marginal costs condition becomes:

$$\alpha + (r_{x,0} - \alpha)\exp\left(\eta \frac{\tau M\left[r_{b,0} + \lambda_b \tau M(1 - s^2)\right]}{\ln M}\right) = r_{b,0} + 2\lambda_b \tau M(1 - s) \tag{A.6}$$

This is the first key equation that determines the equilibrium choice (M, s).

The first order condition respect to M leads to the following condition:

$$\frac{1}{\eta M} = \tau\left[r_{b,0} + 2\lambda_b \tau M(1 - s^2)\right] - \frac{\tau\left[r_{b,0} + \lambda_b \tau M(1 - s^2)\right]}{\ln M} \tag{A.7}$$

This is the second key condition that determines our equilibrium.

Equations (A.6) and (A.7) together determine the equilibrium values of M and s for any given parameters, including the recovery rate α. To analyze how s responds to a change in α, we use the implicit function theorem. First, we define:

$$F^1(M,s,\alpha) = r_{b,0} + 2\lambda_b \tau M(1-s) - \alpha - (r_{x,0} - \alpha)\exp\left(\eta \frac{\tau M\left[r_{b,0} + \lambda_b \tau M(1-s^2)\right]}{\ln M}\right)$$

$$F^2(M,s) = \frac{1}{\eta M} - \tau\left[r_{b,0} + 2\lambda_b \tau M(1-s^2)\right] + \frac{\tau\left[r_{b,0} + \lambda_b \tau M(1-s^2)\right]}{\ln M}$$

These functions are zero in equilibrium. By the implicit function theorem, we have:

$$\begin{pmatrix} \frac{\partial F^1}{\partial M} & \frac{\partial F^1}{\partial s} \\ \frac{\partial F^2}{\partial M} & \frac{\partial F^2}{\partial s} \end{pmatrix} \times \begin{pmatrix} \frac{\partial M}{\partial \alpha} \\ \frac{\partial s}{\partial \alpha} \end{pmatrix} = \begin{pmatrix} -\frac{\partial F^1}{\partial \alpha} \\ -\frac{\partial F^2}{\partial \alpha} \end{pmatrix}$$

Using Cramer's rule and noting that F^2 does not depend directly on α, so $\partial F^2/\partial\alpha = 0$, we have the following derivative of s with respect to α:

$$\frac{ds}{d\alpha} = \frac{\frac{\partial F^1}{\partial \alpha} \times \frac{\partial F^2}{\partial M}}{\frac{\partial F^1}{\partial M} \times \frac{\partial F^2}{\partial s} - \frac{\partial F^1}{\partial s} \times \frac{\partial F^2}{\partial M}}$$

The sign of the numerator is negative. $\frac{\partial F^1}{\partial \alpha} = e^{\eta T} - 1$ is positive, while $\frac{\partial F^2}{\partial M}$ is the second derivative of the profit function with respect to M. If the profit function is concave in M, we can assume that $\frac{\partial F^2}{\partial M} < 0$, which is simply a regularity condition that we assume.

The sign of the denominator is more subtle. Instead of working out the derivatives, we focus on the stability of our equilibrium. Variables move in the direction that increases profit. Since F^1 and F^2 represent the optimality conditions (zero at equilibrium), we analyze the following dynamics:

$$\frac{dM}{dt} = k_1 F^2(M,s), \quad \frac{ds}{dt} = k_2 F^1(M,s,\alpha)$$

with positive constants k_1, k_2 measuring the speed of adjustment. The positive sign of these constants reflect the following: if $F^2 > 0$, then from equation (A.7) the marginal benefit of increasing imports exceeds the marginal cost, so M should rise; if $F^1 > 0$, then from equation (A.6) the marginal cost of domestic bank credit exceeds that of trade credit, so s should rise.

To analyze local stability, we linearize the system around the equilibrium (M^*, s^*), where $F^1(M^*, s^*, \alpha) = 0$ and $F^2(M^*, s^*) = 0$. Let $M(t) = M^* + \Delta M(t)$ and $s(t) = s^* + \Delta s(t)$, where

ΔM and Δs are small deviations. Substituting into the adjustment equations and expanding F^2 and F^1 as Taylor series around (M^*, s^*), we obtain the following linear system:

$$\frac{d}{dt}\begin{pmatrix}\Delta M \\ \Delta s\end{pmatrix} = \begin{pmatrix} k_1 \frac{\partial F^2}{\partial M} & k_1 \frac{\partial F^2}{\partial s} \\ k_2 \frac{\partial F^1}{\partial M} & k_2 \frac{\partial F^1}{\partial s}\end{pmatrix}\begin{pmatrix}\Delta M \\ \Delta s\end{pmatrix} \tag{A.8}$$

Since k_1, k_2 are positive constants, the stability properties of this system are determined by the matrix:

$$A = \begin{pmatrix} \frac{\partial F^2}{\partial M} & \frac{\partial F^2}{\partial s} \\ \frac{\partial F^1}{\partial M} & \frac{\partial F^1}{\partial s}\end{pmatrix} \tag{A.9}$$

A necessary and sufficient condition for the stability of this system is $\mathrm{tr}(A) < 0$ and $\det(A) > 0$. Notice that our denominator in $\partial s/\partial \alpha$ is $-\det(A) < 0$. Then we have:

$$\frac{ds}{d\alpha} = \frac{(\text{positive}) \times (\text{negative})}{(\text{negative})} > 0$$

Thus, under the assumptions of an interior equilibrium, concavity of the profit function in M, and stability of the equilibrium, the trade credit share $s = F/(\tau M)$ is strictly increasing in the recovery rate α.

□

Proposition 2. *The decline in imports induced by an increase in the baseline domestic lending rate $r_{b,0}$ is attenuated when recovery rates on trade credit are higher:*

$$\frac{\partial^2 \ln M}{\partial \alpha \, \partial r_{b,0}} > 0.$$

Proof. The equilibrium is defined by the two conditions

$$F^1(M, s, \alpha, r_{b,0}) = 0, \qquad F^2(M, s, r_{b,0}) = 0$$

By the implicit function theorem, there exist differentiable functions $M(\alpha, r_{b,0})$ and $s(\alpha, r_{b,0})$

satisfying these conditions. Differentiating these functions with respect to a parameter $\theta \in \{\alpha, r_{b,0}\}$ gives:

$$\begin{pmatrix} F_M^1 & F_s^1 \\ F_M^2 & F_s^2 \end{pmatrix} \begin{pmatrix} M_\theta \\ s_\theta \end{pmatrix} = - \begin{pmatrix} F_\theta^1 \\ F_\theta^2 \end{pmatrix}$$

Let J be the determinant of the matrix.

$$J = F_M^1 F_s^2 - F_s^1 F_M^2$$

We apply Cramer's rule to obtain:

$$M_r = -\frac{N}{J}, \qquad M_\alpha = -\frac{F_\alpha^1 F_s^2}{J},$$

where N is as follows:

$$N = F_{r_{b,0}}^1 F_s^2 - F_{r_{b,0}}^2 F_s^1.$$

The elasticity of imports with respect to the bank rate is therefore:

$$\frac{\partial \ln M}{\partial r_{b,0}} = \frac{M_{r_{b,0}}}{M}$$

And differentiating this with respect to α yields:

$$\frac{\partial^2 \ln M}{\partial \alpha \partial r_{b,0}} = -\frac{1}{M^2} M_\alpha M_{r_{b,0}} + \frac{1}{M} \frac{\partial^2 M}{\partial \alpha \partial r_{b,0}}$$

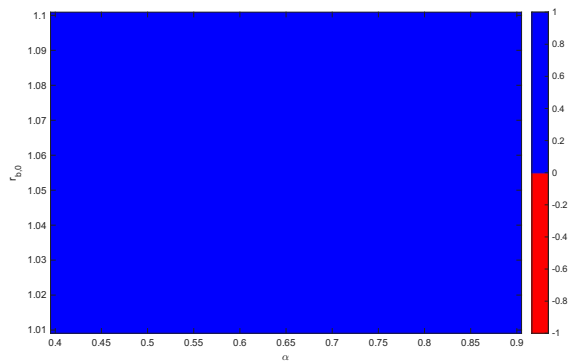
where:

$$\frac{\partial^2 M}{\partial \alpha \partial r_{b,0}} = -\frac{(N_\alpha + N_M M_\alpha + N_s s_\alpha) J - N(J_\alpha + J_M M_\alpha + J_s s_\alpha)}{J^2}$$

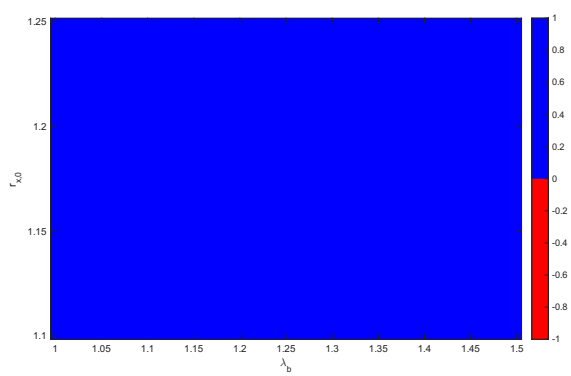
The partial derivatives of N and J involve second derivatives of F^1 and F^2. Substituting all these expressions into (9) gives an explicit, but extremely long, formula for $\frac{\partial^2 \ln M}{\partial \alpha \partial r_{b,0}}$. Thus, we

evaluate the derivative numerically.

Figure A.1 displays the sign of the cross-partial derivative $\frac{\partial^2 \ln M}{\partial \alpha \partial r_{b,0}}$ over two different parameter spaces. Blue regions indicate positive values; red regions indicate negative values. Panel A varies the recovery rate α and the domestic deposit rate $r_{b,0}$ while keeping other parameters fixed. The derivative is positive throughout this space, confirming that a higher recovery rate attenuates the negative impact of a rise in $r_{b,0}$ on imports. Panel B varies the domestic banking cost slope λ_b and the trade credit base rate $r_{x,0}$ with α and $r_{b,0}$ held constant. Again the derivative is uniformly positive.



Panel A: Varying α and $r_{b,0}$ (with $\lambda_b = 1.0$, $r_{x,0} = 1.10$ fixed)



Panel B: Varying λ_b and $r_{x,0}$ (with $\alpha = 0.7$, $r_{b,0} = 1.10$ fixed)

Figure A.1: Numerical evaluation of $\frac{\partial^2 \ln M}{\partial \alpha \partial r_{b,0}}$

This figure shows the cross-partial derivative of lnM with respect to the recovery rate $alpha$ and baseline domestic rate $r_{b,0}$. Blue regions indicate positive values; red regions indicate negative values. In Panel A, we fix $\lambda_b = 1.0$ and $r_{x,0} = 1.10$; in Panel B, we fix $\alpha = 0.7$ and $r_{b,0} = 1.10$. Other parameters are fixed at $\eta = 0.1$ and $\tau = 1.0$.

□

Proposition 3. *Following a negative demand shock in one destination, exporters find it optimal to reallocate a positive quantity toward alternative markets only when the shock size exceeds $\gamma(1+\tau)$. The magnitude of reallocation increases with the shock size and, for a wide range of parameters, decreases with the friction γ.*

Proof. The exporter chooses $\delta \in [0, q_1^*]$ to maximize post-reallocation revenue net of the fixed cost, which is already sunk at the moment of choosing δ. Hence the optimization problem is

$$\max_{\delta \in [0, q_1^*]} R(\delta),$$

where

$$R(\delta) = \big[a_1' - b_1(q_1^* - \delta)\big](q_1^* - \delta) + \big[a_2 - b_2(q_2^* + (1-\gamma)\delta)\big]\big(q_2^* + (1-\gamma)\delta\big).$$

$R(\delta)$ is a quadratic and concave function in δ. Thus, from the first order condition $\frac{dR}{d\delta} = 0$, we have the interior optimum reallocation:

$$\delta^* = \frac{\Delta a - \gamma(1+\tau)}{2\big[b_1 + b_2(1-\gamma)^2\big]}$$

A necessary and sufficient condition for a positive reallocation is $\Delta a > \gamma(1+\tau)$. If $\Delta a \leq \gamma(1+\tau)$, the exporter sets $\delta = 0$. This establishes the first part of the proposition.

The magnitude of reallocation increases strictly with the size of the demand shock, as noticed from differentiating δ^* with respect to Δa:

$$\frac{\partial \delta^*}{\partial \Delta a} = \frac{1}{2\big[b_1 + b_2(1-\gamma)^2\big]} > 0.$$

Finally, differentiating δ^* with respect to γ gives:

$$\frac{\partial \delta^*}{\partial \gamma} = \frac{-(1+\tau)\big[b_1 + b_2(1-\gamma)^2\big] - \big[\Delta a - \gamma(1+\tau)\big] \cdot 2b_2(1-\gamma)(-1)}{2\big[b_1 + b_2(1-\gamma)^2\big]^2}.$$

Simplifying and rearranging the numerator:

$$\frac{\partial \delta^*}{\partial \gamma} = \frac{-(1+\tau) + 4b_2(1-\gamma)\delta^*}{2\big[b_1 + b_2(1-\gamma)^2\big]}.$$

The denominator is always positive. Hence, $\partial\delta^*/\partial\gamma < 0$ iff $4b_2(1-\gamma)\delta^* < 1+\tau$. Since $\delta^* \leq q_1^*$, a sufficient condition is obtained by replacing δ^* with its upper bound q_1^*:

$$4b_2(1-\gamma)q_1^* < 1+\tau.$$

Replacing $q_1^* = \dfrac{a_1 - (1+\tau)}{2b_1}$, and rearranging terms, this condition can be written as:

$$\left(1 + \frac{b_1}{2b_2}\right)(1+\tau) + \gamma\big[a_1 - (1+\tau)\big] - a_1 > 0.$$

Since $a_1 > (1+\tau)$, as otherwise pre-shock sales would be zero, the left-hand side is decreasing in γ. Even for $\gamma = 0$, the condition becomes $\left(1 + \frac{b_1}{2b_2}\right)(1+\tau) > a_1$. This is satisfied whenever the initial demand intercept is not too large relative to the trade cost. For positive γ, the condition is even weaker. Hence, the mild assumption that $a_1 < \left(1 + \frac{b_1}{2b_2}\right)(1+\tau)$ is sufficient to have $\partial\delta^*/\partial\gamma < 0$.

□

Proposition 4. *There exists a unique threshold $\Delta a^* > \gamma(1+\tau)$ such that $G(\Delta a^*) = 0$, with reallocation occurring iff $\Delta a > \Delta a^*$. This threshold increases with the fixed cost f.*

Proof. Recall the gain from reallocation defined in the text:

$$G(\Delta a) = \Pi_{\text{re}}(\Delta a) - \Pi_{\text{no}}(\Delta a),$$

We analyze $G(\Delta a)$ in three regions.

Region 1: $\Delta a \leq \gamma(1+\tau)$. From optimal reallocation choice, $\delta^* = 0$ in this region. Hence $\Pi_{\text{re}}(\Delta a) = R(0) - f = (a_1' - b_1 q_1^*)q_1^* + R_2^* - f$. Since $\Delta a \leq \gamma(1+\tau) < 1+\tau$, we are in the

non-destruction regime where $\Pi_{\text{no}}(\Delta a) = (a_1' - b_1 q_1^*)q_1^* + R_2^*$. Thus

$$G(\Delta a) = -f < 0 \quad \text{for all } \Delta a \leq \gamma(1+\tau). \tag{1}$$

Region 2: $\gamma(1+\tau) < \Delta a \leq 1+\tau$**.** Here $\delta^* > 0$ and $\Pi_{\text{no}}(\Delta a) = (a_1' - b_1 q_1^*)q_1^* + R_2^*$. Applying the envelope theorem to the maximized revenue $R^*(\Delta a)$ gives:

$$\frac{dR^*}{da_1'} = q_1^* - \delta^*.$$

Hence

$$\frac{d\Pi_{\text{re}}}{da_1'} = q_1^* - \delta^*, \qquad \frac{d\Pi_{\text{no}}}{da_1'} = q_1^*.$$

Since $da_1' = -d\Delta a$, we obtain

$$\frac{dG}{d\Delta a} = \delta^* > 0.$$

Thus G is strictly increasing on $(\gamma(1+\tau), 1+\tau]$. At the left endpoint, $\delta^* = 0$, so $G(\gamma(1+\tau)) = -f$. At the right endpoint, $G(1+\tau) = \Pi_{\text{re}}(1+\tau) - \Pi_{\text{no}}(1+\tau)$, which may be positive or negative depending on parameters.

Region 3: $\Delta a > 1+\tau$**.** In this region, the no-reallocation optimum involves destroying part of the production: $\Pi_{\text{no}}(\Delta a) = \dfrac{(a_1')^2}{4b_1} + R_2^*$. For reallocation, δ^*, provided in the main text, imply that $R^*(\Delta a)$, $\Pi_{\text{re}}(\Delta a)$, and thereby $G(\Delta a)$ are continuous functions of Δa.

As $\Delta a \to a_1$ (i.e., $a_1' \to 0$), the no-reallocation profit tends to R_2^*. The reallocation profit is at least the profit from shifting all output to country 2, which is strictly greater than R_2^*, provided $q_2^* + (1-\gamma)q_1^*$ does not exceed the revenue-maximizing quantity. If it does, the exporter can always choose a lower δ to avoid negative marginal revenue. Hence $\lim_{\Delta a \to a_1} G(\Delta a) > 0$.

Summarizing, we have that $G(\Delta a) < 0$ for $\Delta a \leq \gamma(1+\tau)$ and G is strictly increasing on $(\gamma(1+\tau), 1+\tau]$. Then, if $G(1+\tau) > 0$, there exists a unique $\Delta a^* \in (\gamma(1+\tau), 1+\tau]$ such that $G(\Delta a^*) = 0$. On the other hand, if $G(1+\tau) \leq 0$, we have to note that G is continuous and becomes positive for sufficiently large Δa. Thus, the Intermediate Value Theorem guarantees at least one

crossing in $(1+\tau, \infty)$. Since G cannot cross back to negative for $\Delta a > 1+\tau$, there exists a unique $\Delta a^* > \gamma(1+\tau)$ with $G(\Delta a^*) = 0$.

For the last part of the proposition, we can rewrite the gain function as $G(\Delta a) = H(\Delta a) - f$, where $H(\Delta a) = R^*(\Delta a) - \Pi_{\text{no}}(\Delta a)$. The threshold Δa^* satisfies $H(\Delta a^*) = f$. Since H is strictly increasing on $(\gamma(1+\tau), 1+\tau]$ and eventually increasing on $(1+\tau, \infty)$, an increase in f requires a larger Δa to achieve $H(\Delta a) = f$. Hence $\partial \Delta a^* / \partial f > 0$, i.e., the threshold increases with the fixed cost.

$\square$

B Additional Tables and Figures

Table B.1: Effect of Shift in Provisions on firm's debt for both Importers and Exporters, Importers only and Exporters only

Panel A: Bank-firm level regressions			
Dep. Variable: Log(Credit)	$M \cap X$	$M \neg X$	$X \neg M$
	(1)	(2)	(3)
Exposure × *Policy*	-0.033**	-0.026**	-0.028**
	(0.013)	(0.012)	(0.011)
Controls×time	Yes	Yes	Yes
Firm-time FE	Yes	Yes	Yes
Bank-firm FE	Yes	Yes	Yes
Obs.	648,342	314,621	325,044
R^2	0.97	0.97	0.97
Panel B: Firm level regressions			
Dep. Variable: Log(Credit)	$M \cap X$	$M \neg X$	$X \neg M$
	(1)	(2)	(3)
Exposure × *Policy*	-0.022**	-0.038***	-0.010*
	(0.009)	(0.005)	(0.006)
Controls×time	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
$\hat{\alpha}_{f,t}$	Yes	Yes	Yes
Obs.	141,937	86,796	88,059
R^2	0.98	0.98	0.98

This table presents regression results on lending from bank b to firm f (Panel A) and on total firm's debt (Panel B). Column (1) corresponds to the sample of firms that are simultaneously importers and exporters, column (2) corresponds to the sample of importers that are not exporters, whereas column (3) corresponds to the sample of exporters that are not importers. *Exposure* in Panel A is computed as the ratio of construction and real estate lending to total corporate sector lending by bank b as of the end of 2011. In panel B, it is calculated as the weighted average of the exposure of banks lending to firm f, using outstanding credit at the end of 2011 as weights. In Panel A, *Controls* are measured as of the end of 2011 and include bank-firm characteristics (the collateralization rate, the ratio of long-term lending, the share of firm f's NPLs with the bank, and the share of lending coming from bank b out of firm f's total bank lending) and bank characteristics (dummies for quartiles of size, measured by total assets, and quartiles of exposure to importer/exporter companies, measured as the lending share to importers and exporters relative to total corporate sector lending). In Panel B, *Controls* are the weighted average of bank-firm characteristics and bank characteristics, measured as of the end of 2011 and weighted by banks' lending to firm f as of 2011. All variables are defined in Appendix C. We exclude importing and exporting firms in the construction or real estate sectors. Standard errors are double clustered at the bank and firm levels and reported in parentheses. $***$ $p < 0.01$, $**$ $p < 0.05$, $*$ $p < 0.10$.

Table B.2: Effect of Shift in Provisions on Lending to Importing and Exporting Firms, Bank-Firm Level Analysis Accounting for Bank Specialization

Dep. Variable: Log(Credit)	*M*		*X*	
	(1)	(2)	(3)	(4)
Exposure × *Policy*	-0.031**	-0.031**	-0.031**	-0.032***
	(0.013)	(0.012)	(0.012)	(0.011)
Q_2 × Policy		0.014		0.010
		(0.011)		(0.010)
Q_3 × Policy		-0.002		-0.002
		(0.008)		(0.007)
Q_4 × Policy		-0.022**		-0.017**
		(0.008)		(0.008)
Controls×time	Yes	Yes	Yes	Yes
Firm-time FE	Yes	Yes	Yes	Yes
Bank-firm FE	Yes	Yes	Yes	Yes
Obs.	962,963	962,963	973,386	973,386
R^2	0.97	0.97	0.97	0.97

This table reports bank–firm-level regressions of log credit based on equation (12). Columns (1)–(2) correspond to importers and columns (3)–(4) to exporters. *Exposure* is defined as bank b's pre-reform share of lending to construction and real estate (end-2011). In columns (2) and (4), we augment the specification with interactions between *Policy* and indicators for bank specialization in countries from (or to) which the firm imports (or exports), following Paravisini et al. (2023). *Controls* (measured at end-2011) include bank–firm characteristics (collateralization, share of long-term lending, firm f's NPL share at bank b, and bank b's share of the firm's total borrowing) and bank characteristics (size and trade-exposure quartiles). All variables are defined in Appendix C. Firms in construction and real estate are excluded. Standard errors are double-clustered at the bank and firm levels. $***$ $p < 0.01$, $**$ $p < 0.05$, $*$ $p < 0.10$.

Bank Specialization: To ensure that our results reflect the domestic regulatory shock rather than shocks to firms' trade partners, we test whether bank specialization in specific foreign markets drives our findings. Following Paravisini et al. (2023), we augment specification (12) by interacting the *Policy* dummy with indicators for whether a bank specializes in countries from (or to) which the firm imports (or exports). We measure bank b's specialization in country c—for the 50 main export destinations—as: $S_b^c = \frac{\sum_f C_{f,b} \times X_{f,c}}{\sum_j \sum_f C_{f,b} \times X_{f,j}}$, where $C_{f,b}$ is firm f's outstanding credit with bank b in 2011, and $X_{f,c}$ denotes firm f's exports to country c over 2009–2013. The denominator aggregates this term across all destinations. This measure captures the share of a bank's lending portfolio associated with firms exporting to country c. Table B.2 shows that the coefficient on *Exposure* × *Policy* remains stable and statistically significant. Thus, our baseline results are not driven by bank specialization in particular foreign markets, but by the domestic credit supply shock induced by the provisioning reform.

Table B.3: Effect of Shift in Provisions on Lending to Tradable and Non-tradable firms, Bank-Firm Level Analysis

Dep. Variable: Log(Credit)	*All Firms* (1)	$M + X$ (2)	*All Firms* (3)
$Exposure \times Policy$	-0.019**	-0.029**	-0.015*
	(0.009)	(0.012)	(0.009)
$Exposure \times Policy \times Tradable$			-0.011
			(0.008)
Controls×time	Yes	Yes	Yes
Firm-time FE	Yes	Yes	Yes
Bank-firm FE	Yes	Yes	Yes
Obs.	5,513,983	1,288,007	5,513,983
R^2	0.98	0.97	0.98

This table reports bank–firm-level regressions of log credit at the intensive margin based on equation (12). Columns (1) and (3) include all firms (tradable and non-tradable), while column (2) restricts the sample to firms engaged in international trade. *Exposure* is defined as bank b's pre-reform share of lending to construction and real estate (end-2011). *Controls* (measured at end-2011) include bank–firm characteristics (collateralization, long-term lending share, firm f's NPL share at bank b, and bank b's share of the firm's total borrowing) and bank characteristics (size and trade-exposure quartiles). All variables are defined in Appendix C. Firms in construction and real estate are excluded. Standard errors are double-clustered at the bank and firm levels. $***$ $p < 0.01$, $**$ $p < 0.05$, $*$ $p < 0.10$.

Tradable vs. Non-Tradable firms: As an additional robustness check, we examine whether our results are driven by foreign demand shocks disproportionately affecting tradable firms. To this end, we augment specification (12) by interacting the $Exposure \times Policy$ term with an indicator for firms engaged in international trade. Table B.3 reports the results. While the baseline effect of $Exposure \times Policy$ remains negative and statistically significant, the interaction with the tradable indicator is small and statistically insignificant. This indicates that the credit contraction does not differ significantly between tradable and non-tradable firms, suggesting that our findings are unlikely to be driven by shocks to firms' trade partners and instead reflect the domestic regulatory intervention.

Table B.4: Characteristics of Importers and Exporters

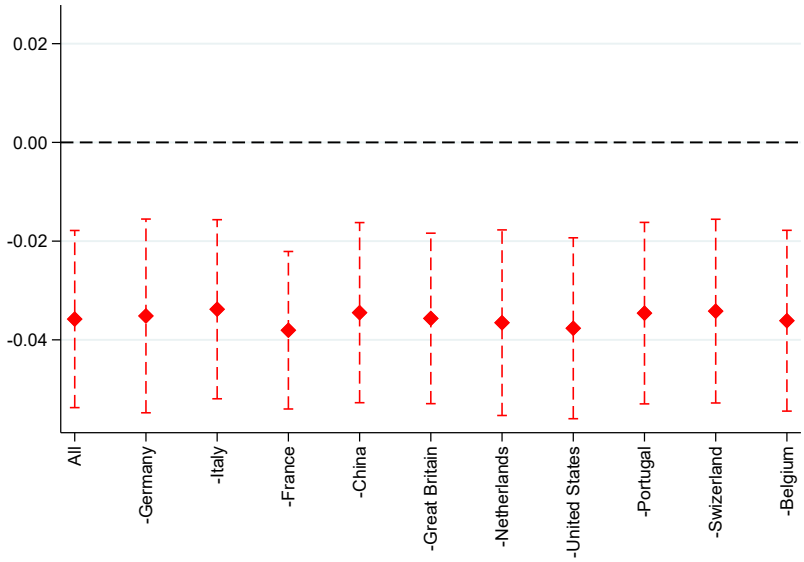
Firm characteristics	**Importers**				**Exporters**			
	p75	p50	p25	mean	p75	p50	p25	mean
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
log(assets)	9.61	8.46	7.49	8.61	9.66	8.54	7.52	8.65
log(employees)	4.04	2.96	1.94	3.03	4.14	3.17	2.09	3.14
log(age)	3.21	2.83	2.30	2.69	3.26	2.83	2.30	2.70
log(debt)	8.40	6.99	5.56	6.90	8.46	7.12	5.65	6.97
log(bank debt)	7.92	6.61	5.15	6.43	7.99	6.74	5.23	6.50
average collateral	0.11	0	0	0.11	0.14	0	0	0.12
Nr. of banks	5	3	2	3.82	5	3	2	3.88
Credit concentration (HHI_{firm})	0.99	0.52	0.30	0.58	0.98	0.51	0.30	0.58
log(tangible fixed assets)	7.91	6.44	4.67	6.23	8.01	6.62	4.92	6.41
log(LT financial investment)	6.48	4.10	2.00	4.33	6.65	4.29	2.13	4.44
log(LT external funds)	7.65	6.28	4.88	6.24	7.70	6.39	5.00	6.32
log(LT funds from financial institutions)	7.24	5.97	4.59	5.89	7.31	6.09	4.71	5.97

This table reports, in columns 1-4, the 75th, 50th, 25th percentile and mean of importer characteristics, and in columns 5-8, the corresponding values for exporters.

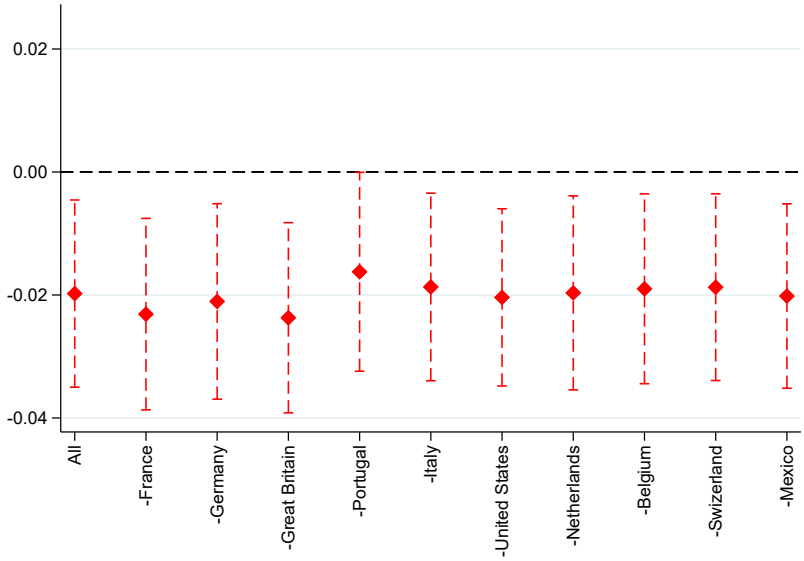
Table B.5: Average difference: exporters only minus importers only

Firm characteristics	Difference between importers and exporters			
	p75	p50	p25	mean
	(1)	(2)	(3)	(4)
log(assets)	0.008	0.002	-0.06**	-0.07***
log(employees)	0.22***	0.29***	0.18***	0.13***
log(age)	0.04***	0	0	0.024**
log(debt)	0.1***	0.11***	0.015	0.032
log(bank debt)	0.13***	0.11***	0.06	0.07*
average collateral	0.07***	0	0	0.02***
Nr. of banks	0	0	–	-0.003
Credit concentration (HHI_{firm})	–	0.004	0.010**	0.005*
log(tangible fixed assets)	0.15***	0.26***	0.35***	0.19***
log(long term financial investment)	0.17**	0.24***	-0.04	0.08
log(long term external funds)	0.09**	0.14***	0.16***	0.1***
log(long term funds from financial institutions)	0.1**	0.12***	0.17***	0.11***

This table reports, in columns 1-3, the difference between the sample of only exporters and only importers, at the 75th, 50th and 25th percentile of their characteristics' distribution. These coefficients are estimated through quantile regressions on a dummy that takes the value of 1 if the firm is classified as exporter only and zero if classified as importer only. Column 4 reports the mean difference in these characteristics for the two groups of firms. The statistical significance of these coefficients is based on the t-test.
$***\ p < 0.01$, $**\ p < 0.05$, $*\ p < 0.10$.



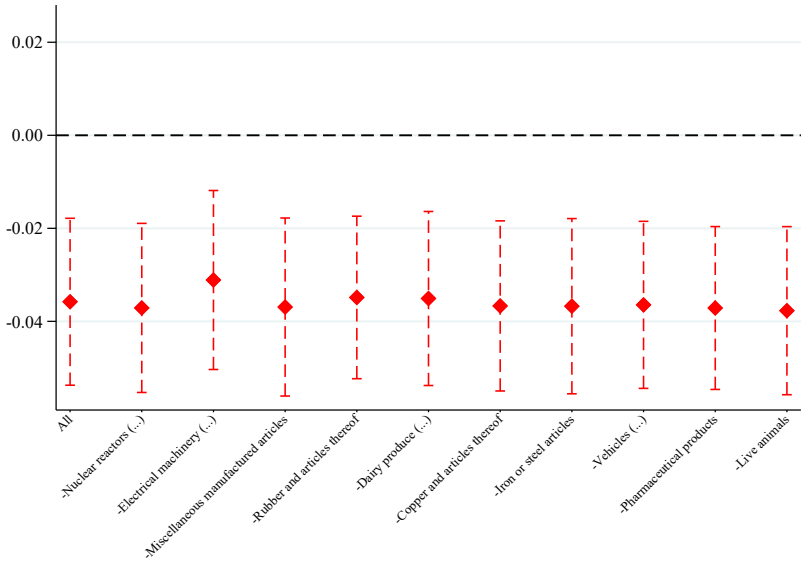
(a) Imports



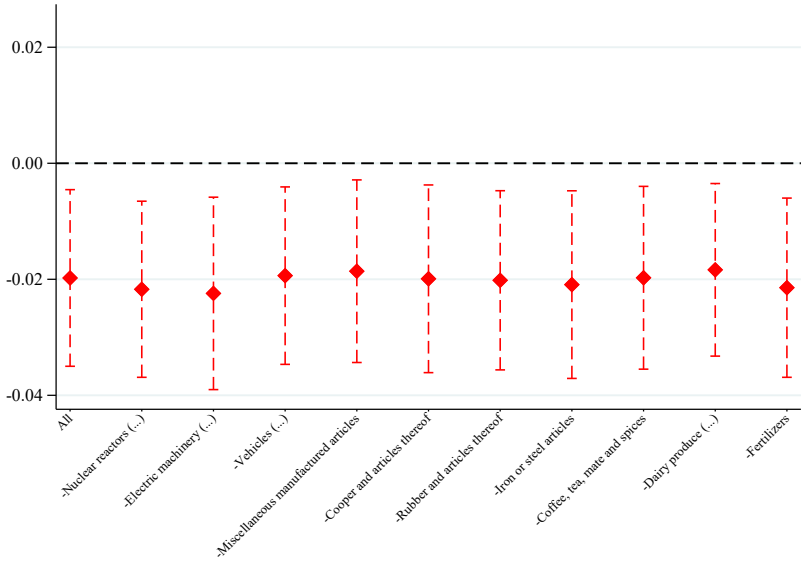
(b) Exports

Figure B.1: Sensitivity Analysis to Main Destination and Origin Countries

The figure presents a sensitivity analysis of our baseline results in specification (14). In Panel (a), we present the sensitivity analysis for imports after sequentially excluding each of the top 10 main origin countries for Spanish imports. In Panel (b), we present the analysis for exports after sequentially excluding each of the top 10 main destinations for Spanish exports. As a benchmark, the first point in each figure (*All*) coincides with the coefficient in columns 1 and 2 of Table 4, respectively. We drop firms in the construction or real estate sectors. Bands represent the 90% confidence intervals. Standard errors are double-clustered at the main bank and firm levels.



(a) Imports



(b) Exports

Figure B.2: Sensitivity Analysis to Main Import and Export Products

The figure presents a sensitivity analysis of our baseline results in specification (14). In Panel (a), we present the sensitivity analysis for imports after sequentially excluding each of the top 10 main imported products for Spanish firms. In Panel (b), we present the analysis for exports after sequentially excluding each of the top 10 main exported products of Spanish firms. As a benchmark, the first point in each figure (*All*) coincides with the coefficient in columns 1 and 2 of Table 4, respectively. We drop firms in the construction or real estate sectors. Bands represent the 90% confidence intervals. Standard errors are double-clustered at the main bank and firm levels.

Table B.6: Impact of Provisioning Shift on Firms' International Trade Flows
Excluding firms highly related to GIIPS

	Mid-point growth		Entry Dummy		Exit Dummy	
	M	X	M	X	M	X
	(1)	(2)	(3)	(4)	(5)	(6)
Exposure × *Policy*	-0.039***	-0.022**	-0.006	-0.000	0.014***	0.010**
	(0.010)	(0.010)	(0.004)	(0.004)	(0.003)	(0.004)
Product-Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Province-Industry-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Product-Country-Firm FE	Yes	Yes	No	No	No	No
Firm FE	No	No	Yes	Yes	Yes	Yes
Obs.	175,864	187,332	177,588	541,365	455,025	230,960
R^2	0.19	0.26	0.25	0.17	0.17	0.17

In this table, we replicate the specification from Table 5, excluding firms highly exposed to GIIPS countries. Exposure is defined using 2006 data–prior to the global financial crisis: we compute the share of each firm's exports and imports originating from or destined to Greece, Italy, Portugal, and Ireland, and exclude firms in the top quartile of this distribution. Standard errors are double-clustered at the main bank and firm levels and are reported in parentheses. $***\ p < 0.01$, $**\ p < 0.05$, $*\ p < 0.10$.

Table B.7: Impact of Provisioning Shift on Firms' International Trade Flows: Excluding Construction-Linked Sectors (IO-Based Robustness)

	Mid-point growth		Entry Dummy		Exit Dummy	
	M	X	M	X	M	X
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Excluding Upstream dependence (Leontief inverse), 90th percentile						
Exposure × *Policy*	-0.033***	-0.019	-0.003	0.002	0.017***	0.009**
	(0.012)	(0.013)	(0.005)	(0.004)	(0.005)	(0.004)
Observations	323490	422205	124159	175176	126038	150649
R^2	0.18	0.19	0.20	0.18	0.24	0.26
Panel B: Excluding Downstream dependence (interior coefficient), 90th percentile						
Exposure × *Policy*	-0.035***	-0.019*	-0.007**	-0.001	0.014***	0.008**
	(0.011)	(0.010)	(0.003)	(0.003)	(0.004)	(0.004)
Observations	552365	637150	211144	269001	219668	225830
R^2	0.17	0.17	0.18	0.16	0.23	0.25
Product-Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Province-Industry-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Product-Country-Firm FE	Yes	Yes	No	No	No	No
Firm FE	No	No	Yes	Yes	Yes	Yes

Notes: This table replicates estimates from Table 5 excluding sectors that are highly linked to construction using Spain's input–output matrix. Upstream dependence is measured using the Leontief inverse $L = (I - A)^{-1}$ and the element $L_{i,\text{construction}}$, capturing the total gross output in sector i required to satisfy one additional unit of final demand in construction. Downstream dependence is measured as the share of domestic intermediate inputs sourced from construction in total intermediate inputs ("interior coefficient"). Panels exclude sectors above the 90th percentile of the corresponding dependence measure. Standard errors are double-clustered at the main bank and firm levels and are reported in parentheses. $***\ p < 0.01$, $**\ p < 0.05$, $*\ p < 0.10$.

Table B.8: Impact of Provisioning Shift on Firms' International Trade Flows, Weighted Regressions

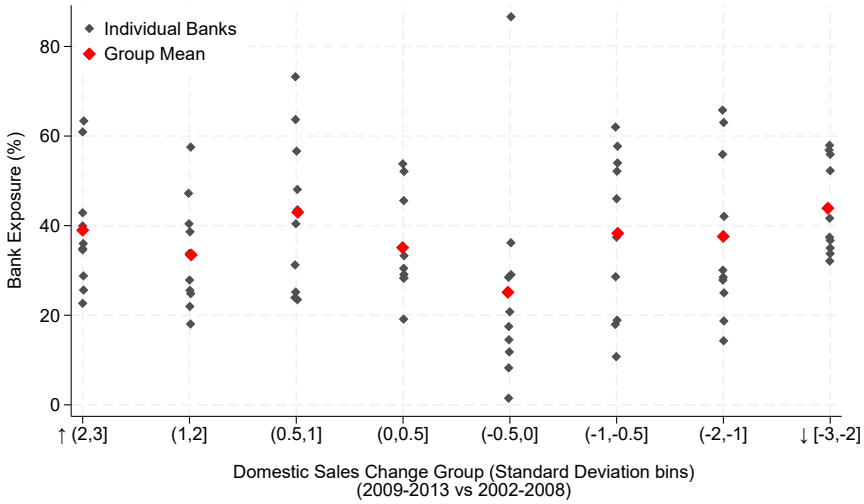
	Mid-point growth			
	Unweighted		Weighted	
	M	X	M	X
	(1)	(2)	(3)	(4)
Exposure × *Policy*	-0.036***	-0.020**	-0.035**	-0.018
	(0.011)	(0.009)	(0.014)	(0.013)
Product-Country-Year FE	Yes	Yes	Yes	Yes
Province-Industry-Year FE	Yes	Yes	Yes	Yes
Product-Country-Firm FE	Yes	Yes	Yes	Yes
Observations	558,540	649,050	558,540	649,050
R-squared	0.17	0.17	0.27	0.27

This table presents the regression estimates on firm's trade flows at the product-country-year level. The dependent variable is the mid-point growth rate between firm f's imports (exports) of product g from (to) country c and the 5-year average of firm f's imports (exports) of the same product from (to) that country, and it is regressed on the firm's exposure and a set of fixed effects. *Exposure* is computed as the average exposure of banks lending to firm f, with weights based on the outstanding credit as of the end of 2011. *Policy* is a dummy variable that takes the value of one after the provisioning increase. The fixed effects included in each regression are noted in the lower part of the table. Columns 1 and 2 present the unweighted results and correspond to columns 1 and 2, respectively, in Table 5. Columns 3 and 4 show the weighted results, using the 5-year average of imported and exported values as weights. Due to the presence of extremely large imported and exported values, weights are winsorized at the 5% level. All variables are defined in Appendix C. Importers and exporters in the construction or real estate sectors are excluded. Standard errors are double-clustered at the main bank and firm levels and are reported in parentheses.
$***\ p < 0.01$, $**\ p < 0.05$, $*\ p < 0.10$.

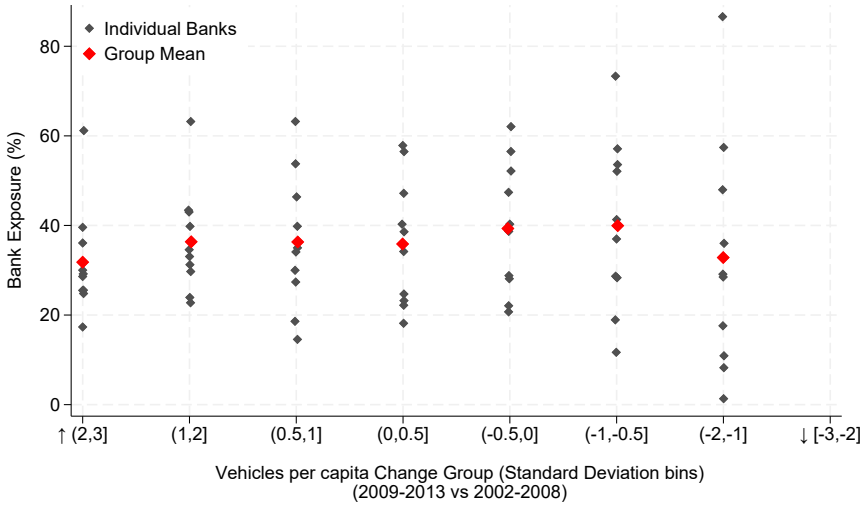
Table B.9: Impact of Provisioning Shift on Firms' International Trade Flows by Cumulative Product-Level Trade Share

	Importers			Exporters		
Product-Level Trade Share:	100%	Bottom 25%	Top 75%	100%	Bottom 25%	Top 75%
	(1)	(2)	(3)	(4)	(5)	(6)
Exposure × *Policy*	-0.036***	-0.035***	-0.035*	-0.020**	-0.022**	-0.011
	(0.011)	(0.010)	(0.020)	(0.009)	(0.011)	(0.016)
Product-Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Province-Industry-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Product-Country-Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	558,540	415,620	135,535	649,050	450,235	188,005
R^2	0.17	0.16	0.28	0.17	0.16	0.26

This table presents the regression estimates on firm's trade flows at the product-country-year level. Columns 1-3 correspond to importers and 4-6 correspond to exporters. Columns 1 and 4 use the full sample of importers and exporters, respectively. Columns 2 and 5 restrict the sample to firms representing the top 75% of total trade value by product in 2011. Columns 3 and 6 restrict the sample to firms representing the bottom 25% of total trade value by product in 2011. *Exposure* is computed as the average exposure of banks lending to firm f, with weights based on the outstanding credit as of the end of 2011. *Policy* is a dummy variable that takes the value of one after the provisioning increase. The fixed effects included in each regression are noted in the lower part of the table. All variables are defined in Appendix C. Importers and exporters in the construction or real estate sectors are excluded. Standard errors are double-clustered at the main bank and firm levels and are reported in parentheses.
$***\ p < 0.01$, $**\ p < 0.05$, $*\ p < 0.10$.



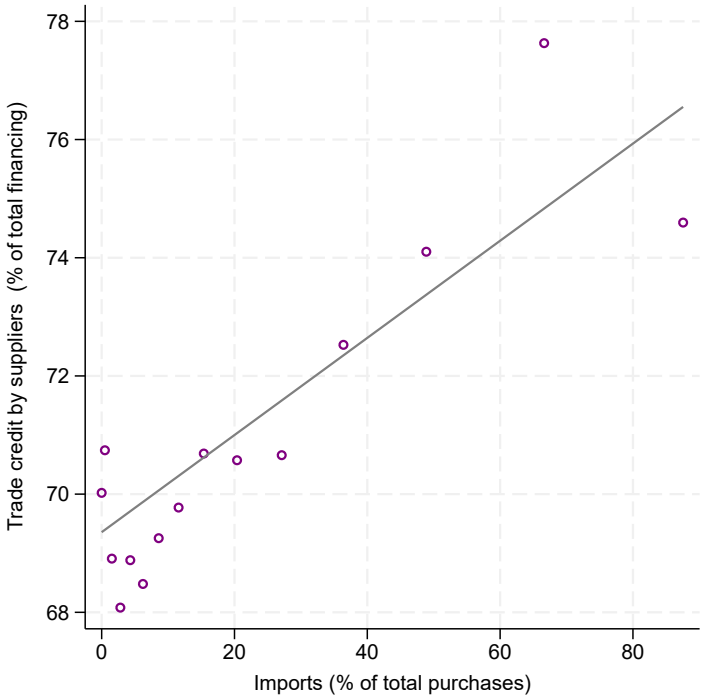
(a) Exposure of Top 10 Lenders by Province Grouped by Change in Domestic Sales



(b) Exposure of Top 10 Lenders by Province Grouped by Change in Vehicles per Capita

Figure B.3: Bank exposure across provinces, based on Almunia et. al (2021)

Panel (a) shows, for each group of provinces on the x-axis–classified based on the standardized percentage change in average firm-level domestic sales between the periods 2002-2008 and 2009-2013–the exposure to the regulation (as defined in Section 5.1) for the top 10 banks with the highest share of loans granted to firms operating in each group. In Panel (b), the groups on the x-axis are defined based on the standardized percentage change in the number of vehicles per capita. The classification of provinces across groups is obtained by reverse-engineering the information presented in Figure 3 of Almunia et al. (2021).



Binscatter of Trade Credit (by suppliers) on Imports (15 bins)

Figure B.4: Trade intensity and use of Trade Credit

This figure presents binned scatterplots illustrating the relationship between firms' trade intensity and their reliance on trade credit. It plots imports as a share of total purchases (x-axis) against trade credit granted by suppliers as a share of total financing (y-axis). Points represent the average values of the X and Y variables within 15 quantile bins of the X-axis variable.

Table B.10: Impact of Provisioning Shift on the Total Exports of Spain's Trade Partners and Trade Costs, Low Financially Developed Origins

	Mid-point growth rate of Exports								
	Far from Spain			Not Spanish Speaking			Far and Not Spanish		
	All	Spain	Rest	All	Spain	Rest	All	Spain	Rest
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$Agg.\ Exposure_{go} \times Post_t$	-0.046*	-0.117***	0.009	-0.022	-0.103***	0.025	-0.065*	-0.181***	-0.003
	(0.027)	(0.039)	(0.020)	(0.023)	(0.032)	(0.018)	(0.033)	(0.045)	(0.025)
Observations	6,651	6,651	6,651	7,231	7,231	7,231	5,611	5,611	5,611
Fixed Effects									
Exporter × Product	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Product charact. × Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country charact. × Year	Yes	Yes	Yes	Yes	Yes	Yes	–	–	–

This table reports regression results for the mid-point growth rate of exports of product g from country o, relative to its five-year average. Estimates consider only low financially developed origins. Columns (1), (4), and (7) present results for total exports to all destinations; columns (2), (5), and (8) for exports to Spain; and columns (3), (6), and (9) for exports to destinations other than Spain. Columns (1)-(3) correspond to exporters that are geographically far from Spain (top three quartiles of the distance-to-Spain distribution). Columns (4)-(6) restrict the sample to exporters from non-Spanish-speaking countries. Columns (7)-(9) restrict the sample to exporters that are both geographically distant and non-Spanish-speaking. The analysis is based on an aggregate measure of market exposure, and incorporates different sets of fixed effects, as specified in equation (19). Aggregate market exposure is calculated as the product-level exposure defined in equation (15) multiplied by Spain's share of country o's exports of product g, following equation (18). The fixed effects included in each regression are listed in the lower part of the table. Product characteristics include a dummy for high or low heterogeneity: i.e, low, if price dispersion of the product is at the bottom quartile of the price dispersion distribution. Exporter country characteristics include: distance to Spain indicator (close to Spain if it belongs to the bottom quartile of the distance-to-Spain distribution), official language (Spanish or non Spanish). All variables are defined in Appendix C. Standard errors shown in parentheses are clustered by product and country of origin.
$***\ p < 0.01$, $**\ p < 0.05$, $*\ p < 0.10$.

C Variable definitions

Firm-bank variables (Source: Credit Register, Bank of Spain)

- *Log of Credit*: The logarithm of committed credit granted by bank b to firm f at quarter t.
- *Termination*: A dummy variable that equals 1 if the bank-firm relationship is terminated in the aftermath of the policy, conditional on its existence as of the end of 2011.
- *Long-term credit ratio*: The ratio of the amount of loans with residual maturity above a year that a firm has with its bank, divided by the credit granted by the bank, as of the end of 2011.
- *Collateralization rate*: The ratio of the amount of collateralized loans that a firm has with its bank, divided by the credit granted by the bank, as of the end of 2011.
- *NPL* ratio: The amount of non-performing loans that a firm has with its bank as of the end of 2011, divided by the credit granted by the bank as of the end of 2011.
- *Credit Share*: The total amount of loans from bank b as of the end of 2011, divided by the firms' total bank debt as of the end of 2011.

Bank variables (Source: Supervisory Reports, Bank of Spain)

- *Exposure*: The amount of loans to the construction and real estate sectors over total lending to non-financial firms as of the end of 2011.
- *Size*: The logarithm of the bank's total assets as of the end of 2011.
- *Capital*: Equity to total assets as of the end of 2011.
- *Liquidity*: Liquid assets (cash and balance with central banks, and loans and advances to governments and credit institutions) to total assets as of the end of 2011.

- *ROA*: Net income to assets as of the end of 2011.
- *NPL ratio*: Non-performing loans as a share of the bank's total credit as of the end of 2011.
- *Rural bank dummy*: A dummy variable that equals 1 if the bank is a rural saving bank.
- *% Lending to X or M*: Lending share to exporters and importers relative to total corporate sector lending as of the end of 2011.
- *Local Govt. Credit to Assets*: Credit to local governments to assets as of the end of 2011.

Firm-level variables (Source: Credit Register, Bank of Spain)

- *Log of Total Bank Credit*: The logarithm of firm's f total bank credit at quarter t.
- $Exposure_f$: Weighted average of the exposure of banks lending to firm f, using the outstanding credit as of the end of 2011 as weights.

Exports and imports, firm-market level (Source: Customs data, Bank of Spain)

- *Exports*: Exports of the 2-digit-HS product g to country c by firm f at year t.
- *Imports*: Imports of the 2-digit-HS product g from country c by firm f at year t.
- *Entry*: A dummy variables that takes the value of one if exports/imports in market (g,c) of firm f are greater than zero at year t, conditional on being zero the previous year.
- *Exit*: A dummy variables that takes the value of one if exports/imports in market (g,c) of firm f are zero at year t, conditional on being positive the previous year.
- *No. Countries*: Number of countries to (from) which a firm exports (imports) in year t.
- *No. Products*: Number of products shipped (purchased) to (from) abroad by a firm in year t.

Firm balance sheet data (Source: Central Balance Sheet Data Office)

- *Total Debt*: Firm's f total debt (bank and non-bank debt) in year t.
- *Employment*: The average number of employees of a firm in year t.
- *Tangible Assets*: Tangible assets of firm f in year t.

Bilateral trade flows (Source: BACI)

- M_{cgot}: Imported value of country c from country o and product g in year t.
- $Exposure_{go}$: Weighted average of banks' exposure to the policy, where the weights account for the relative importance of banks as credit suppliers for firms importing product g from country o.
- *Agg. Exposure*$_{go}$: This aggregate exposure is equal to Exposure$_{go}$ interacted with the share of Spanish purchases in a given market defined by product g and country o.
- X_{got}: Exported value of product g by country o in year t.
- *Comparable countries*: Australia, Belgium, Denmark, Finland, France, Germany, Greece, Ireland, Italy, Japan, Luxembourg, Netherlands, New Zealand, Portugal, Spain, Switzerland and United States.

Country of origin characteristics (Source: CEPII Gravity, Global Financial Development Database, and BankFocus)

- *Close to Spain*: If the distance from the origin country to Spain is below the percentile 25 of the distance to Spain distribution.
- *Spanish Speaking*: If the origin country is a Spanish-speaking country.
- *FTA*: If the origin country has a trade agreement with Spain.
- *Spanish share*: If Spain's share of global purchases for good g is above the median.

- *Price dispersion*: Product's price dispersion is measured as the HS2-level variance of implicit prices, computed at the HS6-digit product level for each country.
- *Heterogeneous products*: If the product's price dispersion is above the percentile 25 of the price dispersion distribution.
- *Financial Development*: If private credit to GDP ratio is above the percentile 75 of the corresponding distribution.
- *Spanish Banks*: If Spanish banks operate in the country.